



ARCHITECTURE

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# STARKWEATHER ACADEMY ADDITIONS & REMODEL

## PLYMOUTH-CANTON COMMUNITY SCHOOLS

### CANTON, MICHIGAN

## PROJECT NUMBER 20069

## CONSTRUCTION DOCUMENTS - BID PACKAGE HS-3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| <p>CONSULTANTS:</p> <p>MECHANICAL ENGINEER</p> <p><b>STRATEGIC ENERGY SOLUTIONS, INC.</b><br/><b>MECHANICAL ELECTRICAL ENGINEERS</b><br/>4000 WEST ELEVEN MILE ROAD<br/>BERKLEY, MICHIGAN 48072<br/>PHONE: (248) 399-1900</p> <p>ELECTRICAL ENGINEER</p> <p><b>STRATEGIC ENERGY SOLUTIONS, INC.</b><br/><b>MECHANICAL ELECTRICAL ENGINEERS</b><br/>4000 WEST ELEVEN MILE ROAD<br/>BERKLEY, MICHIGAN 48072<br/>PHONE: (248) 399-1900</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>KITCHEN EQUIPMENT</p> <p><b>FOOD SERVICE DESIGNS</b><br/><b>KITCHEN EQUIPMENT</b><br/>9201 FUNSTON ST.<br/>WHITE LAKE, MI 48386<br/>PHONE: (248) 366-4513</p> | <p>LIST OF DRAWINGS</p> <p>ARCHITECTURAL</p> <p>TS.1 COMPOSITE FLOOR PLAN<br/>TG.1 GENERAL INFORMATION<br/>LS.1 COMPOSITE LIFE SAFETY PLAN</p> <p>CIVIL</p> <p>C1.1 OVERALL SITE PLAN<br/>C2.1 TOPOGRAPHICAL SURVEY 1 OF 2<br/>C2.2 TOPOGRAPHICAL SURVEY 2 OF 2<br/>C3.1 DEMOLITION PLAN<br/>C4.1 UTILITY PLAN<br/>C5.0 PAVING DETAILS<br/>C5.1 PAVING AND LAYOUT PLAN<br/>C6.1 GRADING PLAN<br/>C7.1 SOIL EROSION AND SEDIMENTATION CONTROL PLAN<br/>S-1 PLYMOUTH TOWNSHIP STANDARD SANITARY SEWER NOTES<br/>S-2 PLYMOUTH TOWNSHIP STANDARD SANITARY SEWER DETAILS</p> | <p>STRUCTURAL</p> <p>S0.1 GENERAL STRUCTURAL NOTES<br/>S0.2 GENERAL STRUCTURAL NOTES (CONT.)<br/>S0.3 SPECIAL INSPECTION SCHEDULES<br/>S0.4 SPECIAL INSPECTION SCHEDULES (CONT.)<br/>S1.1C FOUNDATION PLAN - ZONE 'C'<br/>S1.2C ROOF FRAMING PLAN - ZONE 'C'<br/>S3.1 TYPICAL CONCRETE DETAILS<br/>S4.1 TYPICAL MASONRY DETAILS<br/>S5.1 TYPICAL STEEL DETAILS<br/>S7.1 SECTIONS AND DETAILS<br/>S7.2 SECTIONS AND DETAILS<br/>S7.3 SECTIONS AND DETAILS</p> | <p>ARCHITECTURAL</p> <p>AC.1 COMPOSITE FLOOR PLAN<br/>AD.1 DOOR &amp; 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PIPING<br/>P1.2A FIRST FLOOR PLAN - ZONE 'A' PIPING<br/>P1.2B FIRST FLOOR PLAN - ZONE 'B' PIPING<br/>P1.2C FIRST FLOOR PLAN - ZONE 'C' PIPING<br/>P3.1C ROOF PLAN - ZONE 'C' PLUMBING<br/>P4.0 PLUMBING ENLARGED PLAN<br/>P4.1 PLUMBING ENLARGED PLAN<br/>P4.2 PLUMBING ENLARGED PLAN<br/>P4.3 PLUMBING ENLARGED PLAN<br/>P5.0 PLUMBING DETAILS<br/>P5.1 PLUMBING DETAILS<br/>P6.0 PLUMBING SCHEDULES<br/>M1.1A FIRST FLOOR PLAN - ZONE 'A' MECHANICAL<br/>M1.1B FIRST FLOOR PLAN - ZONE 'B' MECHANICAL<br/>M1.1C FIRST FLOOR PLAN - ZONE 'C' MECHANICAL<br/>M3.1A ROOF PLAN - ZONE 'A' MECHANICAL<br/>M3.1B ROOF PLAN - ZONE 'B' MECHANICAL<br/>M3.1C ROOF PLAN - ZONE 'C' MECHANICAL<br/>W4.0 MECHANICAL ENLARGED PLAN<br/>M5.0 MECHANICAL DETAILS<br/>M5.1 MECHANICAL DETAILS<br/>M6.0 MECHANICAL SCHEDULES<br/>M6.1 MECHANICAL SCHEDULES<br/>M7.0 MECHANICAL DIAGRAMS<br/>M8.0 MECHANICAL TEMPERATURE CONTROLS</p> | <p>ELECTRICAL</p> <p>EO.0 ELECTRICAL GENERAL INFORMATION<br/>EO.1 ELECTRICAL LUMINAIRE SCHEDULE &amp; LIGHTING CONTROLS MATRIX<br/>ED1.1A FIRST FLOOR DEMOLITION - AREA 'A'<br/>ED1.1B FIRST FLOOR DEMOLITION - AREA 'B'<br/>EL1.1A FIRST FLOOR PLAN - ZONE 'A' LIGHTING<br/>EL1.1B FIRST FLOOR PLAN - ZONE 'B' LIGHTING<br/>EL1.1C FIRST FLOOR PLAN - ZONE 'C' LIGHTING<br/>EP1.1A FIRST FLOOR PLAN - ZONE 'A' POWER<br/>EP1.1B FIRST FLOOR PLAN - ZONE 'B' POWER<br/>EP1.1C FIRST FLOOR PLAN - ZONE 'C' POWER<br/>EP3.1A ROOF PLAN - ZONE 'A' POWER<br/>EP3.1B ROOF PLAN - ZONE 'B' POWER<br/>EP3.1C ROOF PLAN - ZONE 'C' POWER<br/>EA.0 POWER ENLARGED PLAN<br/>EO.0 ELECTRICAL DETAILS<br/>EB.0 ELECTRICAL ONE-LINE DIAGRAM<br/>E7.0 ELECTRICAL PANEL SCHEDULES</p> | <p>PROJECT DATA:</p> <p>LOCATION MAP</p> <p>PLYMOUTH, MI</p> <p>NO SCALE</p> <p>ADDRESS: STARKWEATHER ACADEMY<br/>39750 JOY RD.<br/>PLYMOUTH, MI 48170</p>                                                               | <p>BUILDING:</p> <p>BUILDING AREA(S) = 9,584 SQ. FT. (ADDITION)<br/>41,155 SQ. FT. (EXISTING)<br/>50,739 SQ. FT. (TOTAL)</p> <p>CODE:</p> <p>GOVERNING CODES:</p> <ul style="list-style-type: none"><li>- 2016 SCHOOL FIRE SAFETY RULES (2012 Life Safety Code, plus amendments)</li><li>- 2015 MICHIGAN BUILDING CODE</li><li>- 2015 MICHIGAN REHABILITATION CODE FOR EXISTING BUILDINGS</li><li>- 2018 MICHIGAN PLUMBING CODE</li><li>- 2015 MICHIGAN MECHANICAL CODE</li><li>- 2015 MICHIGAN UNIFORM ENERGY CODE (ANSI/ASHRAE/IES Standard 90.1-2013)</li><li>- 2017 MICHIGAN ELECTRICAL CODE (2017 NEC, plus Part 8 Rules)</li><li>- 2010 MICHIGAN ELEVATOR RULES (ASME A17.1-2010, ASME A18.1-2011)</li><li>- MICHIGAN BARRIER FREE CODE (Michigan Building Code 2015 and ICC A117.1-2009)</li><li>- 2013 MICHIGAN BOILER CODE RULES (ASME Boiler and Pressure Vessel Code, 2010 edition, plus 2011a addenda)</li><li>(National Board Inspection Code [NBIC], 2011 edition)</li></ul> <p>CONSTRUCTION CLASSIFICATION:</p> <ul style="list-style-type: none"><li>- EXISTING BUILDING: NFPA TYPE II(000)/MBC TYPE II-B</li><li>- ADDITION: NFPA TYPE II(000)/MBC TYPE II-B</li></ul> <p>USE GROUP CLASSIFICATION : E - EDUCATION<br/>FIRE PROTECTION: NON-SPRINKLERED<br/>01-30-2023 CONSTRUCTION DOCUMENTS<br/>12-14-2022 OWNER REVIEW<br/>10-07-2022 DESIGN DEVELOPMENT<br/>06-20-2022 SCHEMATIC DESIGN</p> <p>DATE ISSUED FOR:</p> |
| <p>LICENSEE'S STATEMENT:</p> <p>This Document has been prepared under the supervision of the Architect, as the person in Responsible Charge with the firm of TMP ARCHITECTURE, INC.. An original embossed or rubber stamp seal and original signature of the Architect is required and shall be affixed to any copy of this Document submitted to a governmental agency for approval or record. This is in conformance with the State of Michigan's PA 299, Article 20 and the General Rules of the Board of Architects.</p> <p>The Architect's seal provided hereon does not take responsibility for certain portions of the Documentation or project requiring the services of a licensed Professional Engineer or other design professional. An original embossed or rubber stamp seal and original signature of the Professional Engineer is required and shall be affixed to any copy of this or other Document submitted to a governmental agency for approval or record. The engineering firms associated with this document are listed above as Consultants.</p> |                                                                                                                                                                  | <p>REGISTRATION SEALS</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>FOOD SERVICE EQUIPMENT</p> <p>FS1 FOOD SERVICE EQUIPMENT PLAN<br/>FS2 FOOD SERVICE UTILITY PLANS</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                         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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>COPYRIGHT</p> <p>© The "architectural work" displayed on these documents is owned exclusively by TMP Architecture, Inc. and may not be used for any purpose without their involvement or express written consent.</p> | <p>PROJECT TITLE</p> <p><b>Starkweather Academy Additions</b></p> <p>PROJECT NO.</p> <p><b>20069</b></p> <p>DRAWING NO.</p> <p><b>TS.1</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |







| DESIGN CRITERIA                                                                                                                                                                                                                                                                           |                                         |                                                        |                   |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------|-------------------|-------------------|
| The structure is designed for the following live loads, in addition to the lateral loads, super-imposed dead loads, and self weight of the structure. Where applicable, the live loads are reduced in accordance with the provisions of the governing Building Code.                      |                                         |                                                        |                   |                   |
| Risk Category                                                                                                                                                                                                                                                                             | III                                     | CODE REFERENCE<br>IBC Table 1604.5<br>ASCE Table 1.5-1 |                   |                   |
| FLOOR LIVE LOADS                                                                                                                                                                                                                                                                          |                                         |                                                        |                   |                   |
|                                                                                                                                                                                                                                                                                           |                                         | CODE REFERENCE                                         |                   |                   |
| Classrooms                                                                                                                                                                                                                                                                                | 40 PSF                                  | ASCE Table 4-1                                         |                   |                   |
| Corridors                                                                                                                                                                                                                                                                                 | 100 PSF                                 | ASCE Table 4-1                                         |                   |                   |
| Assembly                                                                                                                                                                                                                                                                                  | 100 PSF                                 | ASCE Table 4-1                                         |                   |                   |
| Roof                                                                                                                                                                                                                                                                                      | 22 PSF                                  | ASCE Table 4-1                                         |                   |                   |
| SNOW LOADS                                                                                                                                                                                                                                                                                |                                         |                                                        |                   |                   |
|                                                                                                                                                                                                                                                                                           |                                         | CODE REFERENCE                                         |                   |                   |
| Ground Snow Load                                                                                                                                                                                                                                                                          | Pg = 25 PSF                             | ASCE Figure 7-1                                        |                   |                   |
| Flat Roof Snow Load                                                                                                                                                                                                                                                                       | Pl = 25 PSF (minimum)                   | ASCE Section 7.3                                       |                   |                   |
| Exposure Factor                                                                                                                                                                                                                                                                           | Ce = 1.0                                | ASCE Table 7-2                                         |                   |                   |
| Importance Factor                                                                                                                                                                                                                                                                         | I = 1.1                                 | ASCE Table 1.5-2                                       |                   |                   |
| Thermal Factor                                                                                                                                                                                                                                                                            | Ct = 1.0                                | ASCE Table 7.3                                         |                   |                   |
| Snow loads adjacent to vertical projections, on lower roofs adjacent to high roofs, or sloped roofs are increased for the effects of drifting.                                                                                                                                            |                                         |                                                        |                   |                   |
| WIND LOADS                                                                                                                                                                                                                                                                                |                                         |                                                        |                   |                   |
|                                                                                                                                                                                                                                                                                           |                                         | CODE REFERENCE                                         |                   |                   |
| Ultimate Design Wind Speed (3 sec. gust)                                                                                                                                                                                                                                                  | Vult = 120 MPH                          | ASCE Figure 26.5-1C                                    |                   |                   |
| Exposure Category                                                                                                                                                                                                                                                                         | B                                       | ASCE Section 26.7.3                                    |                   |                   |
| Internal Pressure Coefficient                                                                                                                                                                                                                                                             | ± 0.18 (Enclosed)                       | ASCE Section 26.11-1                                   |                   |                   |
| Components and Cladding                                                                                                                                                                                                                                                                   | Per Code Requirements Based on Above    | ASCE Chapter 30                                        |                   |                   |
| ROOF COMPONENTS                                                                                                                                                                                                                                                                           |                                         |                                                        |                   |                   |
|                                                                                                                                                                                                                                                                                           | Zone 1                                  | Zone 2                                                 | Zone 3            | CODE REFERENCE    |
| Support Beams (A > 100 SF)                                                                                                                                                                                                                                                                | -31.8 PSF                               | -37.7 PSF                                              | -37.7 PSF         | ASCE Table 30-7.2 |
| Roof Sheathing (A = 50 SF)                                                                                                                                                                                                                                                                | -32.7 PSF                               | -43.9 PSF                                              | -52.8 PSF         | ASCE Table 30-7.2 |
| Deck Fasteners (A < 10 SF)                                                                                                                                                                                                                                                                | -34.8 PSF                               | -58.3 PSF                                              | -87.8 PSF         | ASCE Table 30-7.2 |
| WALL COMPONENTS                                                                                                                                                                                                                                                                           |                                         |                                                        |                   |                   |
|                                                                                                                                                                                                                                                                                           | Zone 4                                  | Zone 5                                                 | CODE REFERENCE    |                   |
| A = 100 SF                                                                                                                                                                                                                                                                                | +/-29.8 PSF                             | +/-33.1 PSF                                            | ASCE Table 30-7.2 |                   |
| A = 50 SF                                                                                                                                                                                                                                                                                 | +/-31.2 PSF                             | +/-35.9 PSF                                            | ASCE Table 30-7.2 |                   |
| A = 10 SF                                                                                                                                                                                                                                                                                 | +/-34.5 PSF                             | +/-42.4 PSF                                            | ASCE Table 30-7.2 |                   |
| Refer to ASCE 7-10 for zone definitions. Calculate wind load for tributary area of design component. Wind loads shown are for use with ASCE 7-10 load combinations (i.e. 0.6 factor for ASD and 1.0 factor for LRFD), and include directionality factor Kd = 0.85, per ASCE Table 26.6-1. |                                         |                                                        |                   |                   |
| SEISMIC LOADS                                                                                                                                                                                                                                                                             |                                         |                                                        |                   |                   |
|                                                                                                                                                                                                                                                                                           |                                         | CODE REFERENCE                                         |                   |                   |
| Seismic Importance Factor                                                                                                                                                                                                                                                                 | Ie = 1.25                               | ASCE Table 1.5-2                                       |                   |                   |
| Short Period Mapped Spectral Response Acceleration Parameter (5% of Critical Damping)                                                                                                                                                                                                     | SS = 0.0934 g                           | ASCE Section 11.4.1                                    |                   |                   |
| 1.0 sec. Mapped Spectral Response Acceleration Parameter (5% of Critical Damping)                                                                                                                                                                                                         | S1 = 0.0470 g                           | ASCE Section 11.4.1                                    |                   |                   |
| Soil Site Class                                                                                                                                                                                                                                                                           | D                                       | ASCE Section 11.4.2                                    |                   |                   |
| Design Spectral Response Acceleration Parameter (for short period)                                                                                                                                                                                                                        | SDS = 0.100 g                           | ASCE Section 11.4.4                                    |                   |                   |
| Design Spectral Response Acceleration Parameter (1 sec. period)                                                                                                                                                                                                                           | SD1 = 0.075 g                           | ASCE Section 11.4.4                                    |                   |                   |
| Seismic Design Category                                                                                                                                                                                                                                                                   | B                                       | ASCE Section 11.6                                      |                   |                   |
| Seismic Force Resisting System                                                                                                                                                                                                                                                            | Ordinary Reinforced Masonry Shear Walls | ASCE Table 12.2-1                                      |                   |                   |
| Seismic Response Coefficient                                                                                                                                                                                                                                                              | CS = 0.100 Kips                         | ASCE Section 12.8.1.1                                  |                   |                   |
| Response Modification Factor                                                                                                                                                                                                                                                              | R = 2.0                                 | ASCE Table 12.2-1                                      |                   |                   |
| Analysis Procedure                                                                                                                                                                                                                                                                        | Equivalent Lateral Force                | ASCE Section 12.8                                      |                   |                   |
| EARTH PRESSURE LOADS                                                                                                                                                                                                                                                                      |                                         |                                                        |                   |                   |
| LATERAL EARTH EQUIVALENT FLUID PRESSURE                                                                                                                                                                                                                                                   |                                         |                                                        |                   |                   |
| Walls Unbraced at Top                                                                                                                                                                                                                                                                     | 40 PCF                                  |                                                        |                   |                   |
| Walls Braced at Top                                                                                                                                                                                                                                                                       | 55 PCF                                  |                                                        |                   |                   |
| Allowable Soil Bearing Capacity                                                                                                                                                                                                                                                           | 1,500 PSF                               |                                                        |                   |                   |
| 1. Refer to Geotechnical Report for additional Information                                                                                                                                                                                                                                |                                         |                                                        |                   |                   |
| 2. Lateral earth pressure is based upon drained soil. Refer to drawings for foundation drainage.                                                                                                                                                                                          |                                         |                                                        |                   |                   |
| MECHANICAL / ELECTRICAL LOADS                                                                                                                                                                                                                                                             |                                         |                                                        |                   |                   |
| Typical Floors and Roof                                                                                                                                                                                                                                                                   | 10 PSF                                  |                                                        |                   |                   |

| Sheet List   |                                      |                       |                              |  |
|--------------|--------------------------------------|-----------------------|------------------------------|--|
| Sheet Number | Sheet Name                           | Current Revision Date | Current Revision Description |  |
| S0.1         | GENERAL STRUCTURAL NOTES             | 02-20-2023            | ADDENDUM No 1                |  |
| S0.2         | GENERAL STRUCTURAL NOTES (CONT.)     | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S0.3         | SPECIAL INSPECTION SCHEDULES         | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S0.4         | SPECIAL INSPECTION SCHEDULES (CONT.) | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S1.1C        | FOUNDATION PLAN - ZONE C             | 02-20-2023            | ADDENDUM No 1                |  |
| S1.2C        | ROOF FRAMING PLAN - ZONE C           | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S3.1         | TYPICAL CONCRETE DETAILS             | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S4.1         | TYPICAL MASONRY DETAILS              | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S5.1         | TYPICAL STEEL DETAILS                | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S7.1         | SECTIONS AND DETAILS                 | 02-20-2023            | ADDENDUM No 1                |  |
| S7.2         | SECTIONS AND DETAILS                 | 01-30-2023            | CONSTRUCTION DOCUMENTS       |  |
| S7.3         | SECTIONS AND DETAILS                 | 02-20-2023            | ADDENDUM No 1                |  |

#### GENERAL STRUCTURAL NOTES

- The structural notes are intended to augment the drawings and specifications. Should conflicts exist between the Drawings, Specifications and the Structural notes, the strictest provision shall govern.
- The Structural drawings form an integral part of Contract Documents, which include Architectural, Structural, Mechanical, Electrical, Civil/Site drawings and Specifications. Contractor shall coordinate the Structural drawings with the requirements shown in the other components of the Contract Documents.
- Typical details and other sections/details apply to conditions that are similar to the conditions described in the sections/details, even if they are not specifically referenced on the plans.
- The Contractor shall be responsible for means, methods, sequences and procedures of construction.
- The structure is designed to be self-supporting and stable after it is fully completed per requirements of Contract Documents. Contractor shall determine erection procedures and sequence, and ensure the safety of the building and its component parts during erection. This includes the addition of temporary bracing, gusys or tie-downs if necessary. Contractor shall retain ownership of such material after completion of the project.
- Construction shall comply fully with the applicable provisions of OSHA and the local Governing Codes, current edition, and all requirements specified in the codes shall be adhered to as if they were called for or shown on the drawings. This shall not be construed to mean that requirements set forth on the drawing may be modified because they are more stringent than the code requirements or because they are not specifically required by code.
- Governing Building Code – Michigan (International) Building Code 2015.** Standards listed in structural note sections refer to the version and effective date identified in the REFERENCED STANDARDS Chapter in the Governing Building Code.
- Work constructed per these drawings shall be inspected by an Independent Testing Agency retained to ensure compliance with the requirements shown on the Drawings. Special Inspections required by the Governing Building Code, local building department, and the Contract Documents shall be performed by a qualified Special Inspector. Project site visits by the Engineer do not constitute or replace inspection.
- For multi-story Wood construction, flexible joints for architectural, electrical, mechanical, and plumbing work between floors shall be used to eliminate potential issues due to structure movement from wood shrinkage.

#### SHOP DRAWINGS

- Submit shop drawings for review as indicated in material section of general Structural notes.
- Use of Engineering Drawings as erection drawings by the Contractor is strictly prohibited.
- Allow in the schedule - detailing, fabrication and erection - a minimum of 10 working days for review of each shop drawing submitted by the Structural Engineer. Submit shop drawings in reasonable quantities at reasonable intervals (not more than 70 drawings per submittal per week). The 10 working days shall begin when the drawings are received by the Engineer. Review of the shop drawings by the Structural Engineer does not constitute or replace inspection. Submittal per week shall be in addition to the review time required by other project team members. Submit a shop drawing submittal schedule prior to the first submittal.
- Review of shop drawings and other submittals by the Structural Engineer does not relieve the Contractor of the responsibility to check the shop drawings prior to submittal. Errors and omissions associated with the preparation of shop drawings not conforming to the Construction Documents are the responsibility of the shop drawing preparer.
- Shop drawings are an aid for field placement and are superseded by the Contract Documents. Contractor shall ensure that construction is in accordance with the latest Contract Documents. Shop drawing review is only for general compliance with the Contract Documents. Review of the shop drawings by the Structural Engineer does not guarantee that the shop drawings are correct nor infer that the shop drawings supersede the Contract Documents.
- Contractor shall provide a set of approved shop drawings bearing the review stamp of the Structural Engineer, to the local building department and to the project site.
- Notes on submitted shop drawings for work "by others" cannot be responsibility approved by Structural Engineer. Contractor shall coordinate responsibility for materials, connections, etc. prior to shop drawing submittal to the Structural Engineer.
- Contractor shall verify all relevant dimensions and elevations for equipment installations against purchased Manufacturer's certified equipment drawings. Contractor shall coordinate dimensions that depend upon specific equipment, such as elevator openings, mechanical equipment supports, etc., prior to submittal. Such dimensions shall be provided on the shop drawing for the report for additional considerations related to ground water conditions and control, drainage, site preparations, earthwork operations and work adjacent to the existing building.

#### MECHANICAL & ELECTRICAL EQUIPMENT

- Pipes of 3 inches or greater in diameter shall be suspended from steel members, using pipe hangers and clamps. Hangers and clamps shall be aligned with structural member centerlines. One sided beam connections, expansion anchors or other anchors to slabs are not allowed for pipes of 3 inches or greater in diameter. Submit details for pipe hangers and attachments for review by Architect/Structural Engineer.
- Mechanical and electrical equipment weights assumed for structural design are shown on the plans. If the equipment weight varies from that listed, consult with the Architect/Structural Engineer prior to steel shop drawing submittal.

#### SHORING AND BRACING

- Contractor shall provide temporary shoring and bracing of existing construction, new construction, and underground utilities as follows:
  - Where shown or noted on the Drawings.
  - Where existing construction is to be altered or disturbed until permanent support is in place.
  - Where existing construction is not undergoing alteration and is to remain undisturbed but is disturbed as a result of the work of this contract.
  - As required for safe erection, installation of new construction, equipment, etc.
  - When needed for Contractor's "means and methods" of construction and other safety related issues.
- Shoring and bracing shown on the Drawings is conceptual. Contractor shall be responsible for verifying existing conditions, shoring and bracing of existing construction, and transfer of loads through to final load support, and work sequence phasing with new construction.
- Shoring and bracing shall be performed by a Contractor with minimum 5 years demonstrated experience in similar size and scope of shoring and bracing projects.
- Shoring and bracing shall be designed by a Professional Engineer registered in the State of the Project with minimum 5 years demonstrated experience in similar size and scope of shoring and bracing projects. Design loads and methods shall conform to applicable codes. Soil and material strengths shall be verified by tests, unless conservative estimates that do not affect deflections and deformations are approved by the Architect/Structural Engineer.
- Contractor shall submit drawings and calculations sealed and signed by the Contractor's Professional Engineer showing complete design including temporary conditions, final conditions and sequence of work.
- Before starting work, Contractor shall perform condition survey of the existing building structure, exterior facade and interior finishes, including photographic documentation and submit survey to the Owner for record.
- During the shoring and bracing operations, Contractor shall:
  - Keep the existing and new construction in a safe condition.
  - Monitor existing and new construction to detect any signs of distress or deformation.
  - Take immediate steps to prevent distress, deformation or damage.
- Contractor shall continuously monitor the shoring and bracing system. Contractor shall review and ascertain that all field connections are completed according to the Contractor's design and issue approval for inspection of the work by the Testing Agency.
- After completion of shoring and bracing and completion of work requiring shoring and bracing, Contractor shall remove shoring and bracing in a safe and orderly manner, without any cost to the Owner, and to the satisfaction of the Owner and Architect/Structural Engineer.

#### EXISTING CONSTRUCTION

- Contractor shall visit the site and become familiar with the existing conditions.
- Existing building dimensions and conditions shown are based upon original drawings or partial survey and have not been completely field verified. The Owner and Architect/Structural Engineer bear no responsibility for the accuracy of existing dimensions shown. Contractor shall field measure existing dimensions prior to shop drawing preparation and fabrication.
- The analysis of the existing structure is based upon information shown on original drawings by O'Dell, Hewitt & Luckenbach dated 11/30/1996, provided by TMP Architecture.
- Contractor shall verify conditions covering or affecting the structural work, obtain and verify all dimensions and elevations to ensure the proper strength, fit and location of the structural work, report to the Architect/Structural Engineer any and all conditions/discrepancies which may interfere with or otherwise affect or prevent the proper execution and completion of the new work in compliance with the Construction Documents. All discrepancies shall be fully resolved prior to commencing work.
- Existing construction not undergoing alteration is to remain undisturbed. Where such construction is disturbed as a result of the operations of this contract, Contractor shall repair or replace as required and to the satisfaction of the Architect/Structural Engineer and Owner's Representative.

#### EXISTING CONSTRUCTION (CONT.)

- Contractor shall verify the existence, location and elevation of existing utilities, sewers, drains, etc., in demolition areas before proceeding with the work. All discrepancies shall be documented and reported to the Architect/Structural Engineer and Owner's Representative for resolution.
- Should uncharted piping or other utilities be encountered during excavation, Contractor shall consult the Architect/Structural Engineer and Owner's Representative for resolution. Contractor shall provide fire watch during field cutting and welding operations, meeting the Owner's requirements.
- Horizontal wall reinforcement shall be continuous with the minimum lap per ACI 318 unless detailed or noted otherwise. Corner bars shall be provided at changes in wall direction (however small) and shall be of the same size and spacing as the horizontal steel. Each corner bar leg to provide lap splice per ACI 318 unless detailed or noted otherwise. Extend horizontal wall reinforcing through piers.
- Contractor shall provide temporary protection of existing equipment during execution of work, satisfying the Owner's requirements.
- Contractor shall provide temporary protection to prevent damage from the weather and vandalism.
- Contractor shall coordinate work with the Owner's personnel to avoid any interference in their operations.
- Contractor shall coordinate work with the Owner's personnel to avoid any interference in their operations.
- Refer to SHORING AND BRACING notes for additional requirements.

#### FOOTINGS AND FOUNDATIONS

- Contractor shall verify all conditions, including underground utilities and field measurements at job site and report any discrepancies to Owner's Representative.
- Provide necessary sheeting, shoring, bracing, etc. as required during excavations to protect sides of excavations.
- Comply fully with requirements of OSHA and other regulatory agencies for safety provisions.
- Top of spread footing elevations noted on plan are minimum elevations. In all cases, footings are to bear on undisturbed natural soils or engineered fill having a minimum net allowable bearing capacity of 1,500 psf.
- Sides of foundations shall be formed. All concrete surfaces shall be maintained smooth and vertical. Slope sides of excavations as approved by the Geotechnical Engineer, and clean up sloping before and during concrete placement. If existing soil conditions warrant earth forming, Geotechnical Engineer shall make recommendations for specific preparation and procedure to follow.
- Where footing steps are necessary, they shall be no steeper than one vertical to two horizontal unless noted otherwise.
- Footings shall be centered under columns and walls unless specifically detailed otherwise on the Drawings.
- No footings or slabs shall be placed on or against sub-grade containing free water, frost or ice. Should water or frost, however slight, enter a footing excavation after sub-grade approval, the sub-grade shall be re-inspected by the Geotechnical Engineer/Testing Laboratory after removal of water or frost.
- The Contractor shall provide all necessary measures to prevent any frost or ice from penetrating any footing or slab sub-grade before and after placing of concrete until the full building enclosure is completed and heated.
- Excavated material shall be legally disposed of off the Owner's property or stored at the site or used for backfilling operations as required in accordance with the Geotechnical Engineer's recommendations and Project Specification requirements.
- Contractor shall furnish all required de-watering equipment to maintain a dry excavation until backfill is complete.
- Where new footings are adjacent or abut existing foundations, carefully hand excavate and determine bottom of existing foundation. If different than anticipated, adjust new foundations to match existing. In no case shall the new footing be lower than the existing without protection against undermining such as underpinning or shoring.
- Foundation bearing soils shall be inspected by a qualified Geotechnical Engineer. The testing shall include, but not be limited to, identification of soils at and below the foundation bearing level, and the allowable bearing capacity of these soils.
- A Geotechnical Engineer registered in the State of the Project shall inspect the condition and assure the adequacy of all subgrades, fills, backfills before placement of foundations, footings, slabs and walls. They shall submit reports to the Architect/Engineer describing their investigations, including any non-conforming work.
- The design of foundations, retaining walls, and slab on grade is based on the criteria established in the Geotechnical Report No. 223534 by G2 Consulting Group dated October 27, 2022. Refer to the report for additional considerations related to ground water conditions and control, drainage, site preparations, earthwork operations and work adjacent to the existing building.

#### SITE PREPARATION

- Refer to Geotechnical Report No. 223534 by G2 Consulting Group dated October 27, 2022 for consideration related to site preparation and earthwork operations. The requirements and recommendations contained in the report are part of Contract requirements.

#### BACKFILLING

- Where backfill is to be placed on both sides of foundation walls, provide a balanced backfill against foundation walls to eliminate lateral load effects, or provide necessary temporary lateral support to the top of the wall until permanent support is installed.
- Backfill material shall consist of clean, well grade granular soils, free of organic material, silt and clay, or as specified in the Project Specifications.
- Backfill material shall be compacted to 95% of maximum density, as determined by the Modified Proctor Method (ASTM D1557), in lifts not exceeding 6 inches.

#### CAST-IN-PLACE CONCRETE

- Concrete structural framing has been designed by the Ultimate Strength Method per ACI 318 "Building Code Requirements for Structural Concrete".
- Concrete work shall conform to the requirements of ACI 301, "Specifications for Structural Concrete", and ACI 318 "Building Code Requirements for Structural Concrete" except as modified by Structural requirements noted on the Drawings.
- All concrete work shall conform to ACI 201.2R, "Guide to Durable Concrete".
- Cement shall conform to ASTM C150 "Specification for Portland Cement" type I or III.
- Concrete aggregates shall conform to ASTM C33 "Specification for Concrete Aggregates".
- Reinforcing shall conform to ASTM A615 grade 60, unless noted otherwise.
- Reinforcement shall be fabricated and erected according to the ACI standards: "Guide to Presenting Reinforcing Steel Design Details", ACI 315R.
- Welded wire fabric shall be furnished in flat sheets (rolls not permitted) and shall conform to ASTM A1064 and have a minimum size and end lap of 8 inches.
- Welding of reinforcing steel is prohibited unless specifically detailed. Welding where detailed shall conform to AWS D1.4 specification.
- A copy of ACI MNL-15 "Field Reference Manual", ACI 301 "Specifications for Structural Concrete" with selected ACI and ACI 308.1 "Specification for Curing Concrete" for the field office.
- Concrete shall have a minimum 28-day compressive strength as follows:

|               |           |
|---------------|-----------|
| Foundations   | 4,000 psi |
| Slab-on-grade | 3,000 psi |
- Exterior concrete, and interior concrete subjected to freeze/thaw cycles, salt, etc., including walls, shall be air-entrained 6% +/- 1%.
- Concrete shall be normal weight, unless indicated otherwise.
- Contractor shall submit the concrete mix design(s) for review by the Structural Engineer. Proportion mix designs and provide proof of mix design strength as defined in ACI 301. The submittal shall include cement type and source, cement cube strength, aggregate gradations, water tests, admixture catalog information and cylinder strength test results from 30 tests, on specimens with identical mix design, for each concrete mix, or other proof of strength per ACI 301.
- Contractor shall comply with ACI 301 and ACI 306.1 for cold weather concrete placement and shall protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.
- Contractor shall comply with ACI 301 and ACI 305.1 for hot weather concrete placement.
- The approved materials and mix design shall be fully documented and reviewed by the Testing Agency for full compliance. Responsibility for obtaining the required design strength is the Contractor's responsibility.
- Use of calcium chloride, chloride ions, or other salts in concrete is not permitted.
- Samples for strength tests of each class of concrete placed each day shall be taken by the Testing Agency in accordance with Project Specification requirements or ACI 301, whichever is more stringent.
- Contractor shall prepare and submit reinforcement shop drawings to the Structural Engineer for review prior to fabrication. The shop drawings shall clearly show reinforcement lengths and bends, locations of bars, methods of support, details of placement and placement coordination with formwork, embedments, concrete vibration and construction joints. The drawings shall also indicate openings, sleeves, curbs, and concrete dimensions in accordance with ACI 315R. Provide, at minimum wall, column, and beam elevations, wall, column, and beam sections, material schedules, bar lap schedules and locations.

#### CAST-IN-PLACE CONCRETE (CONT.)

- Contractor shall tie reinforcing steel securely in place prior to placing concrete and provide sufficient supports to maintain the position of reinforcing within specified tolerances during all construction activities. Inserting dowels into wet concrete is not permitted.
- Contractor shall provide continuous reinforcement wherever possible; splice only as shown or approved; stagger splices where possible; use splice lengths as noted. Dowels shall match the size and spacing of the specified reinforcement and shall be lapped with tension splices, unless noted otherwise.
- Horizontal wall reinforcement shall be continuous with the minimum lap per ACI 318 unless detailed or noted otherwise. Corner bars shall be provided at changes in wall direction (however small) and shall be of the same size and spacing as the horizontal steel. Each corner bar leg to provide lap splice per ACI 318 unless detailed or noted otherwise. Extend horizontal wall reinforcing through piers.
- Hooked bars shall be standard 90 degree hooks per ACI unless noted otherwise on the Drawings.
- Minimum lap splice shall be Class B per ACI 318 [Increase lap length by 50% for coated bars unless noted otherwise on the drawings]. Location of lap splices shall be as indicated on Construction Documents and/or as shown on the approved reinforcing steel shop drawings.
- Approved rebar couplers may be used to aid placement of dowels through forms. Mechanical splices shall develop 125% of the tensile strength of the rebar.
- Reinforcing steel shall not be cut, bent or straightened in the field unless approved by the Structural Engineer or as indicated on the Drawings.
- Reinforcing steel shall be placed with the following concrete cover and tolerances unless noted otherwise:

|                                                                                                                                                                                                     |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A. Concrete cast against earth (not formed):                                                                                                                                                        | 3"     |
| B. Formed concrete exposed to earth or weather:                                                                                                                                                     |        |
| a. #5 bars or smaller:                                                                                                                                                                              | 1 1/2" |
| b. #6 thru #16 bars:                                                                                                                                                                                | 2"     |
| C. Formed concrete not exposed to earth or weather:                                                                                                                                                 |        |
| a. Slabs, joists, and walls, #11 bars or smaller:                                                                                                                                                   | 3/4"   |
| b. Slabs, joists, and walls, #14 bars or larger:                                                                                                                                                    | 1 1/2" |
| c. Beams, columns, pedestals, and tension ties:                                                                                                                                                     | 1 1/2" |
| D. Clearance between parallel bars in a horizontal layer shall not be less than the bar diameter, 1", or 4/3 d agg, whichever is greater.                                                           |        |
| E. Clearance between parallel bars in two or more horizontal layers, shall not be less than 1" between layers.                                                                                      |        |
| F. Clearance between longitudinal bars in columns, pedestals, struts, and boundary elements in walls shall not be less than 1.5 times the bar diameter, 1 1/2", or 4/3 d agg, whichever is greater. |        |
| G. Maximum deviation from the nominal bar size shall be:                                                                                                                                            |        |
| + .007" for sections with dimensions of 8" or less                                                                                                                                                  |        |
| + 1/2" for sections with dimensions over 8"                                                                                                                                                         |        |
- Tie embeds securely in place prior to placing concrete.
- Do not place pipes or ducts exceeding one quarter the slab or wall thickness within the slab or wall unless specifically shown and detailed on the Structural drawings. Pipes or ducts shall be located within middle third of slab or wall thickness.
- Install inserts and anchors in concrete for suspending mechanical, electrical, and architectural items. If additional fasteners are needed in conventionally reinforced concrete, use drilled-in type anchors located to avoid conflict with reinforcement.
- Anchor rods and tension bolts (furnished by Structural Steel Contractor) shall be set by template to within a 1/8" tolerance in any direction with minimum embedment and exact projection indicated on the drawings, prior to placing concrete.
- No aluminum conduit or products containing aluminum or any other material injurious to the concrete shall be embedded in the concrete.
- Dowels into foundation shall match size and spacing of vertical reinforcement at all columns, piers, and walls, unless noted otherwise.
- Contractor shall coordinate all masonry dowel sizes and spacing to be cast into concrete with masonry reinforcing shop drawings.
- Locate sleeves, openings, embeds, etc. as indicated on the drawings. The Concrete Contractor shall check with other trades to make sure the sleeves, openings, and embeds that are to be provided and set by them are in place prior to placing of concrete in the area involved.
- Contractor shall obtain approval prior to placing openings or sleeves not shown on the Drawings, through any structural member.
- Mortar and grout testing.
- Reinforcement placement and lap verification.
- Verification of clear grout space prior to grouting.
- Verification of proper grouting procedures (grout lift and consolidation).
- Horizontal construction joints are permitted only where indicated. The location of vertical construction joints shall be submitted to the Structural Engineer for review and approval. Construction joints shall be thoroughly mechanically roughened, cleaned and bonding agent applied before placement of adjoining concrete.
- For control joints in slabs, space joints at maximum 15'-0" on center unless noted otherwise on the Drawings. For control joints in walls, space joints at maximum 10'-0" on center unless noted otherwise on the Drawings.
- Construction joints shall be furnished with a full-length keyway centered on members. Where the size of key is not shown on the Drawings, the key shall be 25% of the cross-section dimension of the member and minimum 1 1/2-inches into the first pour of concrete.
- The Concrete Contractor shall be responsible for all pour sequences and construction procedures for all concrete work to account for temperature differentials and shrinkage occurring during the construction phase until the building is permanently in a mechanically controlled environment.
- Coordinate vapor retarder requirements with floor finish requirements.
- Provide pockets or recesses in concrete work for steel columns and beams as required and/or as called for in the Specifications, even if not shown on the Drawings. Provide concrete fill after steel erection.
- Refer to Architectural drawings for slab recesses and for floor finish materials and requirements.
- Provide recess in top of basement walls and grade beams, where applicable, for door openings, ramps, for support of thickened floor slabs, and to receive door jambs.
- Concrete shall be placed to the constant top of slab elevations, while maintaining the minimum concrete thickness noted on the Drawings.
- Construction joints for supported slabs shall be as noted on the Drawing or as approved by the Structural Engineer.
- The use of chlorides such as deicing salts is prohibited for melting ice prior to placement of concrete.
- Sizes of concrete placements shall not exceed the following, unless otherwise indicated on the plans:

|                     |                                                                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. Walls:           | 40 feet maximum length                                                                                                                               |
| B. Slabs on grade:  | Place in alternating strips (approximate width 30 feet & maximum length 200 feet)                                                                    |
| C. Supported slabs: | Place in sections with a maximum area of 12,000 square feet and a maximum length of 100 feet (all concrete slabs including those cast on metal deck) |
- For floor finish tolerances for interior slabs, refer to Specifications.
- Curing of concrete surfaces shall conform to ACI 308.1 "Specification for Curing Concrete" and ACI 308R "Guide to External Curing of Concrete".
- Exterior concrete shall be finished with a broom finish.
- Joints between the structural (and architectural) members shall be properly prepared and filled with joint sealant unless noted otherwise. All joint edges, including top and bottom surfaces and vertical and horizontal surfaces shall be formed or tooled as required. Joint sealant shall be applied only to the top, vertical, and horizontal surfaces unless noted otherwise on the Drawings.
- Joints to be prepared and filled with joint sealant shall include, but are not limited to, construction joints, control joints, isolation joints, and all interface joints between similar and dissimilar members. Specific locations may be indicated on the Drawings, or may be required by approved shop drawings, or may occur due to the construction sequence selected by the Contractor.
- Prior to placing concrete adjacent to existing concrete, mechanically roughen, then thoroughly clean and de-grease existing concrete surfaces. Apply epoxy bonding agent prior to placing fresh concrete. Bonding agent shall be "Sika Armatex 110 EpoCem" by Sika Corporation, or approved equal. Follow all Manufacturer's instructions for surface preparation, mixing, and application.
- Prior to placing concrete topping, mechanically roughen, then thoroughly clean and de-grease existing concrete surfaces. Soak existing concrete surfaces for minimum 12 hours. Place a concrete-slurry of cement and water within 1 hour of topping placement.
- Concrete toppings shall be reinforced with collated, fibrillated, polypropylene fibrous reinforcement.
- Non-shrink grout shall conform to ASTM C1107. Grout shall be premixed, non-shrink, non-catalyzed natural aggregate grout with a minimum 7-day compressive strength of 7,000 psi plastic, 6,000 psi flowable, and 5,000 psi fluid consistency.
- Reinforcing steel, anchor rods, and embed placement shall be inspected, prior to placement of concrete, in accord

#### MASONRY NOTES

- Concrete masonry has been designed in accordance with TMS 402/602, "Building Code Requirements and Specifications for Masonry Structures".
- Concrete Masonry to have a minimum 28-day compressive strength  $f_m = 2,000$  psi unless noted otherwise.
- Concrete Masonry units (CMU) shall conform to the following standards:

|                         |                      |
|-------------------------|----------------------|
| A. Load-Bearing Units:  | ASTM C90             |
| B. Medium Weight Units: | 105 to 125 pcf       |
| C. Normal Weight Units: | greater than 125 pcf |
- Load-bearing CMU shall be at minimum medium weight units, unless noted otherwise.
- Mortar for all masonry shall conform to ASTM C270 with minimum compressive strength of 1,800 psi. Mortar shall be type M. Elsewhere mortar may be either type M or S unless specifically indicated otherwise. Use either Portland cement/mortar or masonry cement for mortar.
- Grout shall conform to ASTM C476 with minimum 28-day compressive strength of 3,000 psi.
- Steel bar reinforcement shall conform to ASTM A615, grade 60.
- Horizontal joint reinforcement shall be "Ladder" type with W17 diameter wire for low walls without cavity wall with veneer and 3/16 inch diameter wires for walls with cavity and veneer. Spacing of horizontal joint reinforcing shall be 16" on center, maximum.
- Minimum vertical CMU wall reinforcing shall be continuous #5 bars at 48" on center, unless noted otherwise.
- Dowels to concrete foundation or slab to match size and spacing of reinforcing unless noted otherwise. Reinforce CMU core at corners, each side of control joints and each side of wall openings with additional (2) #5 continuous vertical reinforcing bars.
- Vertical cells containing reinforcing and grout shall form a continuous cavity, free of mortar droppings.
- Horizontal lintels shall be placed at the top of all masonry wall openings with (2) #5 minimum continuous horizontal reinforcing bars positioned at the bottom of the fully grouted lintel, unless noted otherwise. Coordinate lintel elevations with Architectural Drawings and approved masonry reinforcing shop drawings.
- Horizontal bond beams shall be placed at all floor levels, all stair landing levels, roof level, and top of parapets. Bond beams shall be reinforced with (2) #5 minimum continuous horizontal reinforcing bars positioned at the top of the fully grouted bond beam, unless noted otherwise. Coordinate bond beam elevations with Architectural Drawings and approved masonry reinforcing shop drawings.
- Horizontal bond beam and vertical reinforcing shall be continuous unless noted otherwise. Lap splice reinforcing per the schedule below or use mechanical splices adequate for 125% of specified yield strength of the bar. Lap vertical reinforcing with minimum dowels of same size and spacing that have been previously installed in the foundations. Dowel embedment in concrete shall conform to the requirements of the CAST-IN-PLACE CONCRETE notes.

#### POST-INSTALLED ANCHORS

- Post-installed anchors include all mechanical and adhesive anchors noted on Construction Documents. All post-installed anchors shall conform to ACI 309 for mechanical anchors and AC308 for adhesive anchors.
  - Use only code approved anchors with valid ICC-ESR Evaluation Report for use in base material shown on the Construction Documents. Submit ICC-ESR Evaluation Report to Structural Engineer and Special Inspection Agent for approval.
  - Installer of post-installed anchors shall be trained by anchor Manufacturer.
  - Clean existing concrete surface to solid structural concrete. Grind smooth for full steel contact and to prevent gaps between steel and concrete. Alternatively, provide non-shrink grout in all voids between steel and base material.
  - Drill small diameter pilot hole in existing concrete and check for existing reinforcing. Do not cut or damage existing reinforcing.
  - If existing reinforcing is found, shift hole to avoid existing reinforcing. Submit location of new hole to Structural Engineer for review.
  - Install mechanical anchors and adhesive anchors in strict accordance with Manufacturer's written recommendations and procedure detailed in ICC-ESR Evaluation Report.
  - Special Inspections are required for all mechanical and adhesive anchors. Inspect and test post-installed anchors as specified in ICC-ESR Evaluation Report.
  - Adhesive for rebar and anchors in concrete has been designed based on cracked concrete and seismic applications as applicable, in accordance with ACI 305.4 and ICC-ESR AC308. Design adhesive bond strength shall be based on ACI 355.4 Temperature Category A with installation into dry holes, using a carbide drill bit into cracked concrete that has been cured for at least 21 days.
  - The following anchors are approved. Submittals for alternative embed depths will be reviewed by Structural Engineer and approved at their discretion.
- | Anchor Type: | Approved Anchor | ICC-ESR Report No. | Base Material |
|--------------|-----------------|--------------------|---------------|
|              |                 |                    |               |



STRUCTURAL STEEL

- Design, fabrication and erection of structural steel shall be in accordance with the American Institute of Steel Construction (AISC) 360 Specification for Structural Steel Buildings and the Steel Construction Manual, Load and Resistance Factor Design LRFD.
- Structural steel shall conform to the following ASTM specifications and minimum yield strength:

|                                 |              |                         |
|---------------------------------|--------------|-------------------------|
| W Shapes                        | A572 Gr. 50  | F <sub>y</sub> = 50 ksi |
| Miscellaneous shapes and plates | A36          | F <sub>y</sub> = 36 ksi |
| Round Tubes                     | A500 Grade B | F <sub>y</sub> = 42 ksi |
| Pipe                            | A53 Grade B  | F <sub>y</sub> = 35 ksi |
| Square Tubes                    | A500 Grade B | F <sub>y</sub> = 46 ksi |
- Masonry and brick lintels shall be galvanized G90 per ASTM A123.
- Anchor rods shall conform to ASTM F1554 Grade 36, unless noted Grade 55 or other on Drawings.
- Structural steel bolting shall be ASTM A325 type N, 3/4" diameter snug tight except where other size, ASTM A490 N, pre-tensioned or slip-critical type bolts are indicated.
- ASTM A490 bolts in tension shall be pre-tensioned.
- Shear connectors shall conform to the requirements of "Structural Welding Code – Steel" of the American Welding Society, ANSI/AWS D1.1, F<sub>x</sub> = 65 ksi, as manufactured by Nelson Stud Welding, Div. of TRW, or approved substitute, and welded as per Manufacturer's written instructions.
- Welding shall be done with appropriate E70 series electrodes compatible with the new and existing steel. Welds and welding procedures shall conform to, and welders shall be qualified in accordance with, the "Structural Welding Code - Steel" of the American Welding Society, ANSI/AWS D1.1.
- Where specifically noted as AESS, steel and connections are Architecturally Exposed Structural Steel. Finish steel in compliance with AISC Code of Standard Practice for Steel Buildings and Bridges, Section 10 - Architecturally Exposed Structural Steel.
- Detailing shall be performed using rational engineering design and standard practice in accordance with the Contract Documents. The typical details shown are approximate only and do not indicate the required number of bolts or weld sizes, unless specifically noted.
- Contractor shall submit for review, typical connection details and calculations sealed by a Professional Engineer registered in the State in which the Project is being constructed for proposed connections and for connections not specifically designed and detailed. Follow the details shown where specific connections are detailed.
- Contractor shall submit for review, engineered drawings showing shop fabrication details, field assembly details and erection diagrams for all structural steel. Show at minimum all details included in these Contract Documents with additional erection details as required to completely define the interconnection of structural steel pieces.
- Fabricator shall be AISC Certified or have an AISC equivalent Quality Assurance program as certified by a qualified independent testing agency.
- Anchor rods, base plates and bearing plates shall be located and built into connecting work, pre-set by templates or similar method prior to concrete placement. Plates shall be set in full beds of non-shrink grout.
- Contractor shall reference Architectural drawings for miscellaneous shapes and plates not shown on structural drawings. These items shall be shop welded to the structural framing sections to minimize field welding.
- The length, dimension and connection detail from new structural member to existing structures shall be field verified before fabrication. Field modifications to the fabricated member or connection are not allowed without prior approval by the Structural Engineer. Contractor shall submit sketches or shop drawings detailing proposed modifications for approval.
- Roof openings, unless otherwise noted, to be framed with L5x3x1/4 LLV. Verify size and location of all openings with the trade involved.
- Contractor shall provide L4x4x1/4 seats at column webs where required for support of roof and floor decks. Provide angle outrigger from exterior columns for slab and roof edge plate support.
- Non-composite beam connections shall be capable of supporting minimum 50% of the Maximum Total Uniform Load, AISC Steel Construction Manual, unless specifically noted on the Drawings.
- Beam connections shall be standard AISC approved connections. Extended shear plate connections protruding from column web only approved where beams/girders on either side of column web have equally loaded bays.
- Simple shear connections shall be capable of end rotation as per the requirements of the AISC Specification, Simple Connections, Specification Section J1.2 and Manual Part 10.
- Connections shall be shop welded in accordance with latest AWS Specifications for E70XX electrodes and field bolted with ASTM A325 or ASTM A490 bolts.
- Contractor shall install A325 and A490 bolts in accordance with the "Specification for Structural Joints Using ASTM A325 or A490 Bolts." Snug tight condition shall be achieved using an impact wrench, to bring the connected plies into firm contact, except where noted as slip-critical, pre-tensioned or finger tight.
- Contractor shall provide slip-critical connections at braced frames, moment connections, beams and columns supporting cranes and equipment, mechanical penthouse and elevator room framing and where bolts are in tension.
- Contractor shall provide 3/4" diameter shoulder bolts, double nuts or tack welded nuts finger tight to allow vertical movement with lock washers at slotted connections of wind columns or as noted.
- Where field welding to existing structural steel is indicated, contractor shall thoroughly clean all surfaces to receive weld, removing rust, paint, dirt and other foreign matter in area of work. Provide fire watch protection acceptable to the Owner.
- Beams shall be fabricated with the natural camber up. Provide cambers as indicated on the drawings.
- Stiffener plates and bearing stiffeners are to be provided in pairs.
- Brick relieving angles shall be bolted to plate work as indicated and field welded after alignment to a tolerance of +/- 1/8" from the location specified on the Architectural drawings. The alignment and adjustment shall be done after placement of concrete on floor slabs and after placement of roofing including ballast where applicable.
- Fabrication and erection tolerances for brick frames shall conform to the requirements for Architecturally Exposed Structural Steel (AESS) per AISC 303 Code of Standard Practice for Steel Buildings and Bridges Section 10, with adjustment to meet the brick shelf tolerance noted above unless otherwise specified.
- Secondary steel framing supporting exterior façade shall have connections with minimum 1 inch lateral and 1/2 inch vertical adjustment - allowance each direction from center of attachment point. Contractor shall provide slotted holes and shims as required to provide adjustment.
- Wood blocking shall be fastened to adjacent steel members using minimum 0.177" diameter powder actuated fasteners or equivalent fasteners coordinated with the steel thickness. Install 2 fasteners at 3" minimum spacing across the member spaced along the length at 24" on center.
- Clean steel per SSPC-SP3 and shall receive one shop coat of paint. Omit paint at holes for slip critical type connections. All structural steel to be fireproofed, encased or in contact with concrete, and on top flange of beams receiving shear connectors.
- Steel above the roof and outside the building envelope (exposed to weather) shall be cleaned per SSPC-SP6 and hot dip galvanized. (Finish paint shall be epoxy second coat and acrylic finish coat by same paint manufacturer as prime paint per approved paint system.)
- Contractor shall provide adjustable channel slots on steel columns for masonry ties at 16" on center maximum vertical spacing at ends of all CMU walls which are adjacent to steel column locations. Contractor shall coordinate requirements with steel and masonry contractors.
- Contractor shall control erection procedures and sequences with relation to temperature differentials, especially with respect to structural steel framing into concrete walls, beams or columns.
- Contractor shall provide temporary bracing as required to ensure stability of the structure under full design loads until the permanent bracing is in place. Provide necessary shoring where required during construction.
- The steel frame is self supporting for lateral loads after:
  - Connections, braces and moment frames have been completely welded and bolted.
  - Masonry bearing and shear walls have reached design strength.
  - The roof deck has been properly installed and attached.
- Shop and Field Testing of welds and/or bolts shall follow AISC Minimum Requirements for Inspection.
- Welding shall be inspected by an AWS Certified Welding Inspector (CWI).
- Contractor shall schedule work to allow the above testing requirements to be completed.

STEEL OPEN WEB JOISTS AND JOIST GIRDERS

- Steel open web joists and joist girders shall be designed, fabricated and erected in accordance with latest Steel Joist Institute (SJI) specifications.
- Manufacturers shall be members of the Steel Joist Institute (SJI).
- Where noted, joists and joist girders shall be designed for additional distributed loads, concentrated loads and moments as indicated.
- All roof joist and joist girders exposed to a work floor shall be designed for a single point load at any location on the lower chord. In addition to the specified loads, as follows:

|                                                            |           |
|------------------------------------------------------------|-----------|
| Over manufacturing, storage warehouses, and repair garages | 2,000 lbs |
| All other occupancies                                      | 300 lbs   |
- All roof joist and joist girders over assembly spaces (Auditoriums, Banquet Halls, large Conference Rooms, etc.) shall be designed for a single point load at any location on the lower chord. In addition to the specified loads, as follows:

|                      |          |
|----------------------|----------|
| Over Assembly spaces | 1000 lbs |
|----------------------|----------|
- Contractor shall modify joist and joist girder seats where indicated and as required for end reactions and moments noted.
- Minimum joists bearing on a CMU wall shall be as follows (measured from the face of the wall) unless noted otherwise:

|                  |
|------------------|
| 6" at an 8" wall |
| 8" at a 12" wall |
- Contractor shall provide top chord extensions as shown on Structural or Architectural drawings.
- Contractor shall provide bottom chord extensions at all columns and exterior spandrels as noted. Where moment connections are indicated on drawings, connect bottom chord after dead load is applied. Size connection for loads indicated.
- Contractor shall provide connections for additional members and bracing shown within the joist depth.
- Contractor shall provide bridging meeting the minimum requirements of SJI, including wind uplift considerations, unless detailed otherwise. (Provide bridging at first interior panel point from support for wind uplift.) Horizontal bridging shall be continuous top and bottom, anchored at each end, and welded to each joist. Diagonal bridging shall be bolted to each joist and clamped at the intersection.
- Joist bridging in first joist space adjacent to masonry walls and shorter span steel wide flange beams shall be horizontal, not diagonal bridging.
- Joists and joist girders shall be cambered for deflection due to dead loads or as specifically noted on plan.
- Live load deflection shall not exceed span/360 for joists, special joists, and joist girders.
- Joist and joist girders shall be connected to support in accordance with SJI minimum requirements, or as required to resist loads shown on the drawings.
- Stagger joists as required to achieve minimum bearing length per SJI.
- Contractor shall submit shop drawings with erection plans, details, and loading diagrams for special joists and joist girders and camber for review by Structural Engineer. Shop drawings shall be sealed and signed by a Professional Engineer registered in the State the Project. Provide sealed design calculations for all special joists and joist girders for review by Structural Engineer.
- Joists and joist girders shall receive one coat of shop primer paint except those receiving fireproofing. Refer to Architectural drawings for fireproofing requirements.

STEEL DECK

- Steel decks shall be as noted on drawings, fabricated and erected in accordance with the latest Steel Deck Institute (SDI) specifications.
- Manufacturer shall be a member of the Steel Deck Institute (SDI).
- Steel deck shall have galvanized coating conforming to ASTM A653, coating designation G90 for roof deck and G60 for floor deck. Touch-up paint galvanized surfaces with zinc rich paint after cutting and welding. Clean as required to receive fireproofing. Refer to Architectural drawings for fireproofing requirements.
- Contractor shall provide engineering calculations or published Manufacturer's data and independently certified test data verifying the specified deck requirements. Provide engineered and checked shop drawings indicating location, gage, and size of each piece of decking. Erection drawings shall clearly show details, size, and spacing of connections to structural framing and side laps.
- Steel deck shall be continuous over 3 spans in the direction indicated. Single and double spans, if required, shall satisfy load and deflection requirements.
- Contractor shall provide accessories including closures, "Z" closures, column closures, screed angles, and girder fillers, as required to contain the slab concrete and as required to adequately support the steel deck all sides on the steel framing.
- Steel decking shall be welded as shown or a maximum spacing of 12" on center to the supporting steel with 5/8" diameter puddle welds, unless noted otherwise. For perimeter, moment frame, and braced frame steel members, weld roof deck at a maximum spacing of 6" on center. Side laps shall be fastened at maximum 3'-0" on center, with minimum one fastener per span.
- Steel decking shall be welded to structural steel by qualified welders using pre-qualified procedures. Establish a welding procedure for the plug weld of the steel decking to the structural steel for the particular gage used. Prior to the start of erection of the steel deck, each welder shall be qualified according to AWS requirements.
- No loads shall be permitted to be hung from any roof deck. Mechanical piping over 2 1/2 inches in diameter shall not be hung from floor deck. Hangers for ceilings, ductwork, electrical conduit, piping, etc. shall be directly from structural steel work or supplementary members.

COLD FORMED METAL FRAMING

- All cold formed metal framing members shall be designed, fabricated and erected in accordance with the AISI-S100 "North American Specifications for Design of Cold-formed Steel Structural Members" and in accordance with Manufacturer's written instructions.
- All material shall conform to ASTM A1003, with minimum yield point of 33 ksi for 18 gauge and 50 ksi for 16 gauge and heavier material, and shall have galvanized coating conforming to ASTM A653-G60
- All welding shall conform to "Structural Welding Code – Sheet Steel" of the American Welding Society, AWS D1.3 and AWS D19.0 welding zinc coated steel.
- Unless specifically noted, all material shall be of a minimum metal thickness of 43 mils (18 gauge). Studs serving as backup for brick veneer shall be of a minimum metal thickness of 54 mils (16 gauge).
- All cold formed metal framing members shall meet the deflection requirements of the finish material to be attached to the cold formed metal framing work. Deflection of cold formed metal framing members serving as back up for brick veneer shall not exceed span/600 under serviceability wind load.
- When not specifically designed, the contractor shall submit calculations and layout for stud size, spacing and connection prepared and sealed by a Professional Engineer registered in the State of the Project for review by the Architect/Engineer.
- All studs and joists shall be installed at spacing indicated on the drawings, unless noted, each side of the openings shall be framed with double studs.
- All studs and joists shall have a bridging line installed at a maximum distances of 4'-0" and 5'-0", respectively.
- All Joists shall have web stiffeners at reaction points and concentrated loads.
- The nomenclature used for the design of cold formed metal framing is from the AISI Manual. All members supplied shall meet or exceed the strength shown in the AISI Manual.
- Structural connections of cold formed metal framing members shall be made per manufacturer's recommendations, adequate to carry the imposed loads, and conforming to the AISI and AWS specifications. Connection design to be based on reactions given on the Drawings or as listed in the Manufacturer's uniform loading capacity tables, whichever is greater.



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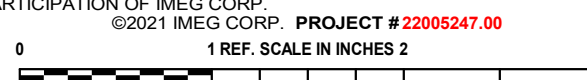
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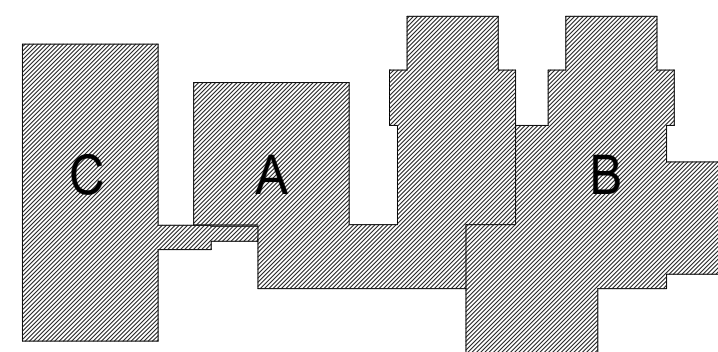
PROJECT TITLE

Starkweather Academy - Addition & Remodeling

Plymouth Canton Community Schools

DRAWING TITLE

GENERAL STRUCTURAL NOTES (CONT.)



Key Plan: NTS

ISSUE DATES

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PROJECT NO.

20069

DRAWING NO.

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| STATEMENT OF SPECIAL INSPECTIONS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
| 1.                               | SPECIAL INSPECTIONS SHALL BE PERFORMED IN ACCORDANCE WITH THE 2018 MICHIGAN (INTERNATIONAL) BUILDING CODE CHAPTER 17 AND AS MODIFIED HEREIN.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |  |
| 2.                               | DESIGNATIONS:<br>SI SPECIAL INSPECTOR QUALIFIED WITH DEMONSTRATED COMPETENCE DOCUMENTED BY CERTIFICATIONS FROM RECOGNIZED AGENCIES SUCH AS AWS, ACI, MASONRY INSTITUTE OF MICHIGAN (MIM), ETC. AS SUBMITTED AND APPROVED BY THE BUILDING OFFICIAL. SPECIAL INSPECTOR MAY BE A FIRM WITH MULTIPLE SPECIALISTS AND A PROJECT MANAGER PROVIDING REPORTS.<br>TA TESTING AGENCY QUALIFIED TO TEST AND INSPECT MATERIALS AND ASSEMBLIES. TESTING AGENCY SHALL BE UNDER THE SUPERVISION OF THE SPECIAL INSPECTOR.<br>GE GEOTECHNICAL ENGINEER WHO PROVIDED THE ORIGINAL PROJECT GEOTECHNICAL SOILS INVESTIGATION REPORT.<br>SE SPECIALTY ENGINEER RESPONSIBLE FOR DESIGNING ASSEMBLIES SUCH AS PRECAST CONCRETE, STEEL JOISTS, COLD FORMED FRAMING ASSEMBLIES, ETC. SPECIALTY ENGINEER SHALL PROVIDE OBSERVATION OF FABRICATED AND INSTALLED ITEMS OF THEIR DESIGN IN ADDITION TO THE SPECIAL INSPECTION.<br>3. TA, GE, AND SE SHALL SUBMIT RECORDS OF THE INSPECTION RESULTS TO THE SI. THE SI SHALL COMPILE AND SUBMIT INSPECTION RECORDS TO THE ARCHITECT/ENGINEER AND BUILDING OFFICIAL. RECORDS SHALL INCLUDE STATEMENTS OF TESTS, WHETHER INSTALLED/FABRICATED ITEM COMPLIES WITH CONTRACT DOCUMENTS, REMEDIAL WORK PERFORMED, RETESTS.<br>4. SI SHALL PROVIDE A DAILY REPORT OF ANY DISCREPANCIES FROM THE CONTRACT DOCUMENTS FOUND ON THE SAME DAY OF THE INSPECTION TO THE ENGINEER OF RECORD. FORMAL REPORTS OF COMPLIANCE CAN FOLLOW BY A MAXIMUM OF 2 WEEKS. SI SHALL PROVIDE AND SIGN FINAL REPORT WITH A SUMMARY OF ALL TESTS PERFORMED AND RESULTS TO THE ENGINEER OF RECORD AND BUILDING OFFICIAL, IN ACCORDANCE WITH SECTION 1704.2.4.<br>5. SI, TA & GE SHALL BE ENGAGED BY THE OWNER IN COMPLIANCE WITH THE MICHIGAN (INTERNATIONAL) BUILDING CODE.<br>6. WHERE FABRICATION OF STRUCTURAL LOAD-BEARING, OR LATERAL LOAD-RESISTING MEMBERS OR ASSEMBLIES IS BEING CONDUCTED ON THE PREMISES OF A FABRICATOR'S SHOP, SPECIAL INSPECTIONS OF THE FABRICATED ITEMS SHALL BE PERFORMED DURING FABRICATION. SPECIAL INSPECTIONS DURING FABRICATION ARE NOT REQUIRED WHERE THE FABRICATOR MAINTAINS APPROVED DETAILED FABRICATION AND QUALITY CONTROL PROCEDURES THAT PROVIDE A BASIS FOR CONTROL OF THE WORKMANSHIP AND THE FABRICATOR'S ABILITY TO CONFORM TO APPROVED CONSTRUCTION DOCUMENTS AND THE GOVERNING BUILDING CODE. APPROVAL SHALL BE BASED UPON REVIEW OF FABRICATION AND QUALITY CONTROL PROCEDURES AND PERIODIC INSPECTION OF FABRICATION PRACTICES BY THE BUILDING OFFICIAL. SPECIAL INSPECTIONS ARE NOT REQUIRED WHERE THE FABRICATOR IS REGISTERED AND APPROVED IN ACCORDANCE WITH SECTION 1704.2.5.1.<br>7. REFER TO SPECIAL INSPECTION SCHEDULES AND GENERAL STRUCTURAL NOTES FOR ADDITIONAL QUALITY CONTROL TESTING AND INSPECTIONS. |  |  |  |  |

| SPECIAL INSPECTION REQUIREMENTS - SOILS AND FOUNDATIONS                                                              |                      |          |                     |               |                   |
|----------------------------------------------------------------------------------------------------------------------|----------------------|----------|---------------------|---------------|-------------------|
| INSPECTION TASK                                                                                                      | INSPECTION FREQUENCY |          | REFERENCED STANDARD | IBC REFERENCE | RESPONSIBLE AGENT |
|                                                                                                                      | CONTINUOUS           | PERIODIC |                     |               |                   |
| 1. VERIFY MATERIALS BELOW SHALLOW FOUNDATIONS ARE ADEQUATE TO ACHIEVE THE DESIGN BEARING CAPACITY.                   | -                    | X        | GEOTECHNICAL REPORT | 1705.6        | SI/GE             |
| 2. VERIFY EXCAVATIONS ARE EXTENDED TO PROPER DEPTH AND HAVE REACHED PROPER MATERIAL.                                 | -                    | X        | GEOTECHNICAL REPORT | 1705.6        | SI/GE             |
| 3. PERFORM CLASSIFICATION AND TESTING OF COMPACTED FILL MATERIALS.                                                   | -                    | X        | GEOTECHNICAL REPORT | 1705.6        | SI/GE/TA          |
| 4. VERIFY USE OF PROPER MATERIALS, DENSITIES AND LIFT THICKNESSES DURING PLACEMENT AND COMPACTION OF COMPACTED FILL. | X                    | -        | GEOTECHNICAL REPORT | 1705.6        | SI/GE/TA          |
| 5. PRIOR TO PLACEMENT OF COMPACTED FILL, INSPECT SUBGRADE AND VERIFY THAT SITE HAS BEEN PREPARED PROPERLY.           | -                    | X        | GEOTECHNICAL REPORT | 1705.6        | SI/GE/TA          |

| SPECIAL INSPECTION REQUIREMENTS - CONCRETE CONSTRUCTION                                                                                                                                                                                                 |                      |             |                                               |                                |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------|-----------------------------------------------|--------------------------------|-------------------|
| INSPECTION TASK                                                                                                                                                                                                                                         | INSPECTION FREQUENCY |             | REFERENCED STANDARD                           | IBC REFERENCE                  | RESPONSIBLE AGENT |
|                                                                                                                                                                                                                                                         | CONTINUOUS           | PERIODIC    |                                               |                                |                   |
| 1. INSPECT REINFORCEMENT, INCLUDING PRESTRESSING TENDONS, AND VERIFY PLACEMENT.                                                                                                                                                                         | -                    | X           | ACI 318: Ch. 20, 25.2, 25.3, 26.6.1-26.6.3    | 1908.4                         | SI                |
| 2. REINFORCING BAR WELDING:<br>A. VERIFY WELDABILITY OF REINFORCING BARS OTHER THAN ASTM A706.<br>B. INSPECT SINGLE-PASS FILLET WELDS, MAXIMUM 5/16".<br>C. INSPECT ALL OTHER WELDS.                                                                    | -<br>-<br>X          | X<br>X<br>- | AWS D1.4<br>ACI 318: 26.6.4                   | -                              | SI                |
| 3. INSPECT ANCHORS CAST IN CONCRETE.                                                                                                                                                                                                                    | -                    | X           | ACI 318: 17.8.2                               | -                              | SI / TA           |
| 4. INSPECT ANCHORS POST-INSTALLED IN HARDENED CONCRETE MEMBERS.<br>A. ADHESIVE ANCHORS INSTALLED IN HORIZONTALLY OR UPWARDLY INCLINED ORIENTATIONS TO RESIST SUSTAINED TENSION LOADS.<br>B. MECHANICAL ANCHORS AND ADHESIVE ANCHORS NOT DEFINED IN 4.A. | X<br>-               | -<br>X      | ACI 318: 17.8.2.4<br>ACI318: 17.8.2           | -                              | SI / TA           |
| 5. VERIFY USE OF REQUIRED DESIGN MIX.                                                                                                                                                                                                                   | -                    | X           | ACI 318: Ch. 19, 26.4.3, 26.4.4               | 1904.1, 1904.2, 1908.2, 1908.3 | SI / TA           |
| 6. PRIOR TO CONCRETE PLACEMENT, FABRICATE SPECIMENS FOR STRENGTH TESTS, PERFORM SLUMP AND AIR CONTENT TESTS, AND DETERMINE THE TEMPERATURE OF THE CONCRETE.                                                                                             | X                    | -           | ASTM C172<br>ASTM C31<br>ACI 318: 26.5, 26.12 | 1908.10                        | SI / TA           |
| 7. INSPECT CONCRETE AND SHOTCRETE PLACEMENT FOR PROPER APPLICATION TECHNIQUES.                                                                                                                                                                          | X                    | -           | ACI 318: 26.5                                 | 1908.6, 1908.7, 1908.8         | SI                |
| 8. VERIFY MAINTENANCE OF SPECIFIED CURING TEMPERATURE AND TECHNIQUES.                                                                                                                                                                                   | -                    | X           | ACI 318: 26.5.3-26.5.5                        | 1908.9                         | SI                |
| 9. VERIFY IN-SITU CONCRETE STRENGTH, PRIOR TO STRESSING OF TENDONS IN POST-TENSIONED CONCRETE AND PRIOR TO REMOVAL OF SHORES AND FORMS FROM BEAMS AND STRUCTURAL SLABS.                                                                                 | -                    | X           | ACI 318: 26.11.2                              | -                              | SI / SE / TA      |
| 10. INSPECT FORMWORK FOR SHAPE, LOCATION, AND DIMENSIONS OF THE CONCRETE MEMBER BEING FORMED.                                                                                                                                                           | -                    | X           | ACI 318: 26.11.1.2(b)                         | -                              | SI / SE / TA      |

| SPECIAL INSPECTION REQUIREMENTS - MASONRY: MINIMUM SPECIAL INSPECTION REQUIREMENTS                                                                    |           |         |         |                    |                                 |                                                |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|---------|--------------------|---------------------------------|------------------------------------------------|-------------------|
| MINIMUM SPECIAL INSPECTION                                                                                                                            |           |         |         |                    |                                 |                                                |                   |
| INSPECTION TASK                                                                                                                                       | FREQUENCY |         |         | REFERENCE CRITERIA |                                 |                                                | RESPONSIBLE AGENT |
|                                                                                                                                                       | LEVEL 1   | LEVEL 2 | LEVEL 3 | IBC SECTION        | TMS 402/ ACI 530/ ASCE 5        | TMS 602/ ACI 530.1/ ASCE 6                     |                   |
| 1. AS MASONRY CONSTRUCTION BEGINS, VERIFY THAT THE...                                                                                                 |           |         |         |                    |                                 |                                                | SI                |
| A. PROPORTIONS OF SITE-PREPARED MORTAR.                                                                                                               | NR        | P       | P       | -                  | -                               | ART. 2.1, 2.6 ...                              |                   |
| B. GRADE AND SIZE OF PRESTRESSING TENDONS AND ANCHORAGES.                                                                                             | NR        | P       | P       | -                  | -                               | ART. 2.4 B & 2.4 H                             |                   |
| C. GRADE, TYPE AND SIZE OF REINFORCEMENT, CONNECTORS, ANCHOR BOLTS, AND PRESTRESSING TENDONS AND...                                                   | NR        | P       | P       | -                  | -                               | ART. 3.4 & 3.6 A                               |                   |
| D. PRESTRESSING TECHNIQUE.                                                                                                                            | NR        | P       | P       | -                  | -                               | ART. 3.6 B                                     |                   |
| E. PROPERTIES OF THIN-BED MORTAR FOR AAC MASONRY.                                                                                                     | NR        | C/P     | C       | -                  | -                               | ART. 2.1 C.1                                   |                   |
| F. SAMPLE PANEL CONSTRUCTION.                                                                                                                         | NR        | P       | C       | -                  | -                               | ART. 1.6 D                                     | SI                |
| 2. PRIOR TO GROUTING, VERIFY THAT THE FOLLOWING ARE IN COMPLIANCE:                                                                                    |           |         |         |                    |                                 |                                                |                   |
| A. GROUT SPACE.                                                                                                                                       | NR        | P       | C       | -                  | -                               | ART. 3.2 D & 3.2 F                             |                   |
| B. PLACEMENT OF PRESTRESSING TENDONS AND...                                                                                                           | NR        | P       | P       | -                  | SEC. 10.8 &...                  | ART. 2.4 & 3.6                                 |                   |
| C. PLACEMENT OF REINFORCEMENT, CONNECTORS, AND ANCHOR BOLTS.                                                                                          | NR        | P       | C       | -                  | SEC. 6.1, 6.3.1, 6.3.6, & 6.3.7 | ART. 3.2 E & 3.4                               | SI                |
| D. PROPORTIONS OF SITE-PREPARED GROUT AND...                                                                                                          | NR        | P       | P       | -                  | -                               | ART. 2.6 B & 2...                              |                   |
| 3. VERIFY COMPLIANCE OF THE FOLLOWING DURING...                                                                                                       |           |         |         |                    |                                 |                                                |                   |
| A. MATERIALS AND PROCEDURES WITH THE APPROVED...                                                                                                      | NR        | P       | P       | -                  | -                               | ART. 1.5                                       |                   |
| B. PLACEMENT OF MASONRY UNITS AND MORTAR JOINT CONSTRUCTION.                                                                                          | NR        | P       | P       | -                  | -                               | ART. 3.3 B                                     | SI                |
| C. SIZE AND LOCATION OF STRUCTURAL MEMBERS.                                                                                                           | NR        | P       | P       | -                  | -                               | ART. 3.3 F                                     |                   |
| D. TYPE, SIZE, AND LOCATION OF ANCHORS, INCLUDING OTHER DETAILS OF ANCHORAGE OF MASONRY TO STRUCTURAL MEMBERS, FRAMES, OR OTHER...                    | NR        | P       | C       | -                  | SEC. 1.2.1(e), 6.2.1, & 6.3.1   |                                                |                   |
| E. WELDING OF REINFORCEMENT.                                                                                                                          | NR        | C       | C       | -                  | SEC. 6.1.6.1.2                  |                                                |                   |
| F. PREPARATION, CONSTRUCTION, AND PROTECTION OF MASONRY DURING COLD WEATHER (TEMPERATURE BELOW 40°F (4.4°C)) OR HOT WEATHER (TEMPERATURE ABOVE 90°... | NR        | P       | P       | -                  | -                               | ART. 1.8 C & 1.8 D                             | SI                |
| G. APPLICATION AND MEASUREMENT OF PRESTRESSING...                                                                                                     | NR        | C       | C       | -                  | -                               | ART. 3.6 B                                     |                   |
| H. PLACEMENT OF GROUT AND PRESTRESSING GROUT FOR BONDED TENDONS IS IN COMPLIANCE.                                                                     | NR        | C       | C       | -                  | -                               | ART. 3.5 & 3.6 C                               |                   |
| I. PLACEMENT OF AAC MASONRY UNITS AND CONSTRUCTION OF THIN-BED MORTAR JOINTS.                                                                         | NR        | C/P     | C       | -                  | -                               | ART. 3.3 B.9 & 3.3 F.1.b                       |                   |
| 4. OBSERVE PREPARATION OF GROUT SPECIMENS, MORTAR SPECIMENS, AND/OR PRISMS.                                                                           | NR        | P       | C       | -                  | -                               | ART. 1.4 B 2.a.3, 1.4 B.2.b.3, 1.4 B.2.c.3,... | SI                |
| IBC REFERENCE SECTION 1705.4, AND TMS 602/ACI 530.1/ASCE 6 TABLE 4.                                                                                   |           |         |         |                    |                                 |                                                |                   |

| SPECIAL INSPECTION REQUIREMENTS - POST-INSTALLED ANCHORS                                                                         |                         |               |                   |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------|-------------------|
| INSPECTION TASK                                                                                                                  | REFERENCED STANDARD     | IBC REFERENCE | RESPONSIBLE AGENT |
| 1. INSPECT AND TEST ALL POST-INSTALLED MECHANICAL AND ADHESIVE ANCHORS MANUFACTURER'S ICC-ESR EVALUATION REPORT FOR EACH ANCHOR. | ICC-ESR FOR EACH ANCHOR | 1705.1.1      | SI / TA           |



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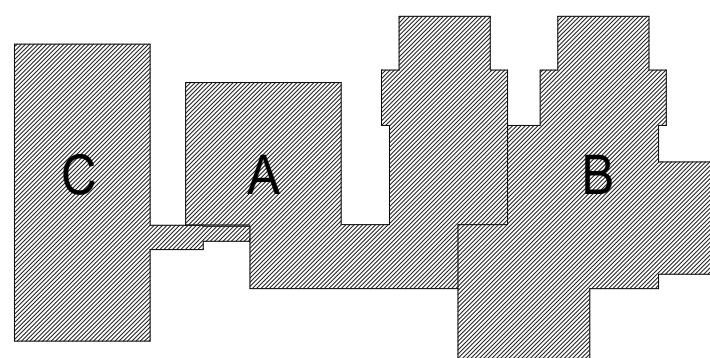
PROJECT TITLE

Starkweather Academy - Addition & Remodeling

Plymouth Canton Community Schools

DRAWING TITLE

SPECIAL INSPECTION SCHEDULES



Key Plan: NTS

ISSUE DATES

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DRAWING NO.

S0.3



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| SPECIAL INSPECTION REQUIREMENTS - STRUCTURAL STEEL                                                                                                                                                                                                                                      |                      |          |                                    |               |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------|------------------------------------|---------------|-------------------|
| INSPECTION TASK                                                                                                                                                                                                                                                                         | INSPECTION FREQUENCY |          | REFERENCED STANDARD                | IBC REFERENCE | RESPONSIBLE AGENT |
|                                                                                                                                                                                                                                                                                         | CONTINUOUS           | PERIODIC |                                    |               |                   |
| 1. INSPECTION OF STEEL FABRICATED ITEMS SHALL BE PERFORMED ON PREMISES DURING FABRICATION.                                                                                                                                                                                              | -                    | X        |                                    |               |                   |
| A. EXCEPTIONS: SPECIAL INSPECTIONS DURING FABRICATION NOT REQUIRED WHERE THE FABRICATOR IS REGISTERED AND APPROVED IN ACCORDANCE WITH SECTION 1704.2.5.1.                                                                                                                               | -                    | -        | AISC QUALITY CERTIFICATION         | 1704.2.5      | SI                |
| 2. SPECIAL INSPECTIONS AND NONDESTRUCTIVE TESTING OF STRUCTURAL STEEL ELEMENTS IN BUILDINGS, STRUCTURES, AND PORTIONS THEREOF SHALL BE IN ACCORDANCE WITH THE QUALITY ASSURANCE INSPECTION REQUIREMENTS OF AIS...                                                                       | X                    | X        | AISC QUALITY CERTIFICATION         | 1705.2.1      | SI                |
| A. SPECIAL INSPECTION OF RAILING SYSTEMS COMPOSED OF STRUCTURAL STEEL ELEMENTS SHALL BE LIMITED TO WELDING INSPECTION OF WELDS AT THE BASE OF CANTILEVERED RAIL POSTS.                                                                                                                  | -                    | X        | AISC QUALITY CERTIFICATION         | 1705.2.1      | SI                |
| QUALITY CONTROL (QC) SHALL BE PROVIDED BY THE FABRICATOR AND ERECTOR. QUALITY ASSURANCE (QA) SHALL BE PROVIDED BY OTHERS WHEN REQUIRED BY THE AUTHORITY HAVING JURISDICTION, APPLICABLE BUILDING CODE, PURCHASER, OWNER, OR ENGINEER OF RECORD.                                         |                      |          |                                    |               |                   |
| INSPECTION TASK                                                                                                                                                                                                                                                                         | INSPECTION FREQUENCY |          | REFERENCED STANDARD                | IBC REFERENCE | RESPONSIBLE AGENT |
|                                                                                                                                                                                                                                                                                         | QC                   | QA       |                                    |               |                   |
| INSPECTION OF BOLTING                                                                                                                                                                                                                                                                   |                      |          |                                    |               |                   |
| 1. INSPECTION TASKS PRIOR TO BOLTING:                                                                                                                                                                                                                                                   |                      |          |                                    |               |                   |
| A. MANUFACTURER'S CERTIFICATIONS AVAILABLE FOR FASTENER...                                                                                                                                                                                                                              | O                    | P        |                                    |               |                   |
| B. FASTENERS MARKED IN ACCORDANCE WITH ASTM REQUIREMENTS.                                                                                                                                                                                                                               | O                    | O        |                                    |               |                   |
| C. CORRECT FASTENERS SELECTED FOR THE JOINT DETAIL (GRADE, TYPE, BOLT LENGTH IF THREADS ARE TO BE EXCLUDED FROM SHEAR PLANE).                                                                                                                                                           | O                    | O        |                                    |               |                   |
| D. CORRECT BOLTING PROCEDURE SELECTED FOR JOINT DETAIL.                                                                                                                                                                                                                                 | O                    | O        | AISC 360, SECTION N5, TABLE N5.6-1 | 1705.2        | SITA              |
| E. CONNECTING ELEMENTS, INCLUDING THE APPROPRIATE FAYING SURFACE CONDITION AND HOLE PREPARATION, IF SPECIFIED, MEET APPLICABLE REQUIREMENTS.                                                                                                                                            | O                    | O        |                                    |               |                   |
| F. PRE-INSTALLATION VERIFICATION TESTING BY INSTALLATION PERSONNEL OBSERVED AND DOCUMENTED FOR FASTENER ASSEMBLIES AND METHODS USED.                                                                                                                                                    | P                    | O        |                                    |               |                   |
| G. PROTECTED STORAGE PROVIDED FOR BOLTS, NUTS, WASHERS AND OTHER FASTENER COMPONENTS.                                                                                                                                                                                                   | O                    | O        |                                    |               |                   |
| 2. INSPECTION TASKS DURING BOLTING:                                                                                                                                                                                                                                                     |                      |          |                                    |               |                   |
| A. FASTENER ASSEMBLIES PLACED IN ALL HOLES AND WASHERS AND NUTS ARE POSITIONED AS REQUIRED.                                                                                                                                                                                             | O                    | O        |                                    |               |                   |
| B. JOINT BROUGHT TO THE SNUG-TIGHT CONDITION PRIOR TO THE PRETENSIONING OPERATION.                                                                                                                                                                                                      | O                    | O        | AISC 360, SECTION N5, TABLE N5.6-2 | 1705.2        | SITA              |
| C. FASTENER COMPONENT NOT TURNED BY THE WRENCH PREVENTING FROM ROTATING.                                                                                                                                                                                                                | O                    | O        |                                    |               |                   |
| D. FASTENERS ARE PRETENSIONED IN ACCORDANCE WITH THE RCSC SPECIFICATION, PROGRESSING SYSTEMATICALLY FROM THE MOST RIGID POINT TOWARD THE FREE EDGES.                                                                                                                                    | O                    | O        |                                    |               |                   |
| 3. INSPECTION TASKS AFTER BOLTING:                                                                                                                                                                                                                                                      |                      |          | AISC 360, SECTION N5, TABLE N5.6-3 | 1705.2        | SITA              |
| A. DOCUMENT ACCEPTANCE OR REJECTION OF BOLTED CONNECTIONS.                                                                                                                                                                                                                              | P                    | P        |                                    |               |                   |
| O. OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS.<br>P. PERFORM THESE TASKS FOR EACH BOLTED CONNECTION.                                                                                                                               |                      |          |                                    |               |                   |
| INSPECTION OF WELDING:                                                                                                                                                                                                                                                                  |                      |          |                                    |               |                   |
| 1. INSPECTION TASKS PRIOR TO WELDING:                                                                                                                                                                                                                                                   |                      |          |                                    |               |                   |
| A. WELDER QUALIFICATION RECORDS AND CONTINUITY RECORDS.                                                                                                                                                                                                                                 | P                    | O        |                                    |               |                   |
| B. WELDING PROCEDURE SPECIFICATIONS (WPSs) AVAILABLE.                                                                                                                                                                                                                                   | P                    | P        |                                    |               |                   |
| C. MANUFACTURER CERTIFICATIONS FOR WELDING CONSUMABLES...                                                                                                                                                                                                                               | P                    | P        |                                    |               |                   |
| D. MATERIAL IDENTIFICATION (TYPE/GRADE).                                                                                                                                                                                                                                                | O                    | O        |                                    |               |                   |
| E. WELDER IDENTIFICATION SYSTEM.                                                                                                                                                                                                                                                        | O                    | O        |                                    |               |                   |
| F. FIT-UP OF GROOVE WELDS (INCLUDING JOINT GEOMETRY):<br>- JOINT PREPARATIONS<br>- DIMENSIONS (ALIGNMENT, ROOT OPENING, ROOT FACE, BEVEL)<br>- CLEANLINESS (CONDITION OF STEEL SURFACES)<br>- TACKING (TACK WELD QUALITY AND LOCATION)<br>- BACKING TYPE AND FIT (IF AVAILABLE)         | O                    | O        | AISC 360, SECTION N5, TABLE N5.4-1 | 1705.2        | SITA              |
| G. FIT-UP OF CJP GROOVE WELDS OF HSS, T-, Y-, AND K-JOINTS WITHOUT BACKING (INCLUDING JOINT GEOMETRY):<br>- JOINT PREPARATIONS<br>- DIMENSIONS (ALIGNMENT, ROOT OPENING, ROOT FACE, BEVEL)<br>- CLEANLINESS (CONDITION OF STEEL SURFACES)<br>- TACKING (TACK WELD QUALITY AND LOCATION) | P                    | O        |                                    |               |                   |
| H. CONFIGURATION AND FINISH OF ACCESS HOLES.                                                                                                                                                                                                                                            | O                    | O        |                                    |               |                   |
| I. FIT-UP OF FILLET WELDS:<br>- DIMENSIONS (ALIGNMENT, GAPS AT ROOT)<br>- CLEANLINESS (CONDITION OF STEEL SURFACES)<br>- TACKING (TACK WELD QUALITY AND LOCATION)                                                                                                                       | O                    | O        |                                    |               |                   |
| J. CHECK WELDING EQUIPMENT.                                                                                                                                                                                                                                                             | O                    | -        |                                    |               |                   |
| 2. INSPECTION TASKS DURING WELDING:                                                                                                                                                                                                                                                     |                      |          |                                    |               |                   |
| A. CONTROL AND HANDLING OF WELDING CONSUMABLES:<br>- PACKAGING<br>- EXPOSURE CONTROL                                                                                                                                                                                                    | O                    | O        |                                    |               |                   |
| B. NO WELDING OVER CRACKED TACK WELDS.                                                                                                                                                                                                                                                  | O                    | O        |                                    |               |                   |
| C. ENVIRONMENTAL CONDITIONS:<br>- WIND SPEED WITHIN LIMITS<br>- PRECIPITATION AND TEMPERATURE                                                                                                                                                                                           | O                    | O        |                                    |               |                   |
| D. WPS FOLLOWED:<br>- SETTINGS ON WELDING EQUIPMENT<br>- TRAVEL SPEED<br>- SELECTED WELDING MATERIALS<br>- SHIELDING GAS TYPE/FLOW RATE<br>- PREHEAT APPLIED<br>- INTERPASS TEMPERATURE MAINTAINED (MIN./MAX.)<br>- PROPER POSITION (F, V, H, OH)                                       | O                    | O        | AISC 360, SECTION N5, TABLE N5.4-2 | 1705.2        | SITA              |
| E. WELDING TECHNIQUES:<br>- INTERPASS AND FINAL CLEANING<br>- EACH PASS WITHIN PROFILE LIMITATIONS<br>- EACH PASS MEETS QUALITY REQUIREMENTS                                                                                                                                            | O                    | O        |                                    |               |                   |
| F. PLACEMENT AND INSTALLATION OF STEEL HEADED STUD ANCHORS.                                                                                                                                                                                                                             | P                    | P        |                                    |               |                   |
| 3. INSPECTION TASKS AFTER WELDING:                                                                                                                                                                                                                                                      |                      |          |                                    |               |                   |
| A. WELDS CLEANED.                                                                                                                                                                                                                                                                       | O                    | O        |                                    |               |                   |
| B. SIZE, LENGTH AND LOCATION OF WELDS.                                                                                                                                                                                                                                                  | P                    | P        |                                    |               |                   |
| C. WELDS MEET VISUAL ACCEPTANCE CRITERIA:<br>- CRACK PROHIBITION<br>- WELD/BASE-METAL FUSION<br>- CRATER CROSS SECTION<br>- WELD PROFILES<br>- WELD SIZE<br>- UNDERCUT<br>- POROSITY.                                                                                                   | P                    | P        |                                    |               |                   |
| D. ARC STRIKES.                                                                                                                                                                                                                                                                         | P                    | P        | AISC 360, SECTION N5, TABLE N5.4-3 | 1705.2        | SITA              |
| E. K-AREA.                                                                                                                                                                                                                                                                              | P                    | P        |                                    |               |                   |
| F. WELD ACCESS HOLES IN ROLLED HEAVY SHAPES AND BUILT-UP HEAVY SHAPES.                                                                                                                                                                                                                  | P                    | P        |                                    |               |                   |
| G. BACKING REMOVED AND WELD TABS REMOVED (IF REQUIRED).                                                                                                                                                                                                                                 | P                    | P        |                                    |               |                   |
| H. REPAIR ACTIVITIES.                                                                                                                                                                                                                                                                   | P                    | P        |                                    |               |                   |
| I. DOCUMENT ACCEPTANCE OR REJECTION OF WELDED JOINT OR...                                                                                                                                                                                                                               | P                    | P        |                                    |               |                   |
| J. NO PROHIBITED WELDS HAVE BEEN ADDED WITHOUT THE APPROVAL OF THE EOR.                                                                                                                                                                                                                 | O                    | O        |                                    |               |                   |
| O. OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS.<br>P. PERFORM THESE TASKS FOR EACH WELDED JOINT OR MEMBER.                                                                                                                          |                      |          |                                    |               |                   |

| SPECIAL INSPECTION REQUIREMENTS - OPEN-WEB STEEL JOISTS AND JOIST GIRDERS  |                      |          |                                             |               |                   |
|----------------------------------------------------------------------------|----------------------|----------|---------------------------------------------|---------------|-------------------|
| INSPECTION TASK                                                            | INSPECTION FREQUENCY |          | REFERENCED STANDARD                         | IBC REFERENCE | RESPONSIBLE AGENT |
|                                                                            | CONTINUOUS           | PERIODIC |                                             |               |                   |
| 1. INSTALLATION OF OPEN-WEB STEEL JOISTS AND JOIST GIRDERS:                |                      |          |                                             |               |                   |
| A. END CONNECTIONS - WELDED OR BOLTED.                                     | -                    | X        | SJI SPECIFICATIONS LISTED IN SECTION 2207.1 |               |                   |
| B. BRIDGING - HORIZONTAL OR DIAGONAL.                                      | -                    | -        | -                                           | 1705.2.3      | SI / TA           |
| 1. STANDARD BRIDGING.                                                      | -                    | X        | SJI SPECIFICATIONS LISTED IN SECTION 2207.1 |               |                   |
| 2. BRIDGING THAT DIFFERS FROM SJI SPECIFICATIONS LISTED IN SECTION 2207.1. | -                    | X        | -                                           |               |                   |

| SPECIAL INSPECTION REQUIREMENTS - STEEL CONSTRUCTION OTHER THAN STRUCTURAL STEEL                                                                                               |                      |          |                                 |               |                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------|---------------------------------|---------------|-------------------|
| INSPECTION TASK                                                                                                                                                                | INSPECTION FREQUENCY |          | REFERENCED STANDARD             | IBC REFERENCE | RESPONSIBLE AGENT |
|                                                                                                                                                                                | CONTINUOUS           | PERIODIC |                                 |               |                   |
| 1. MATERIAL VERIFICATION OF COLD-FORMED STEEL DECK:                                                                                                                            |                      |          |                                 |               |                   |
| A. IDENTIFICATION MARKINGS TO CONFORM TO ASTM STANDARDS SPECIFIED IN THE APPROVED CONSTRUCTION DOCUMENTS.                                                                      | -                    | X        | APPLICABLE ASTM MATERIAL...     | 1705.2.2      |                   |
| B. MANUFACTURER'S CERTIFIED TEST REPORTS.                                                                                                                                      | -                    | X        | -                               | -             |                   |
| 2. INSPECTION OF WELDING:                                                                                                                                                      |                      |          |                                 |               |                   |
| A. COLD-FORMED STEEL DECK:                                                                                                                                                     |                      |          |                                 |               |                   |
| 1. FLOOR AND ROOF DECK WELDS.                                                                                                                                                  | -                    | X        | AWS D1.3                        | 1705.2.2      |                   |
| B. REINFORCING STEEL:                                                                                                                                                          |                      |          |                                 |               |                   |
| 1. VERIFICATION OF WELDABILITY OF REINFORCING STEEL OTHER THAN ASTM A706.                                                                                                      | -                    | X        |                                 |               |                   |
| 2. REINFORCING STEEL RESISTING FLEXURAL AND AXIAL FORCES IN INTERMEDIATE AND SPECIAL MOMENT FRAMES, AND BOUNDARY ELEMENTS OF SPECIAL STRUCTURAL WALLS OF CONCRETE AND SHEAR... | X                    | -        | AWS D14, ACI 318: SECTION 3.5.2 | 1705.2.2      | SI / TA           |
| 3. SHEAR REINFORCEMENT.                                                                                                                                                        | X                    | -        |                                 |               |                   |
| 4. OTHER REINFORCING STEEL.                                                                                                                                                    | -                    | X        |                                 |               |                   |

| SPECIAL INSPECTION REQUIREMENTS - COLD FORMED METAL FRAMING                                                                                                                                   |                      |          |                     |               |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------|---------------------|---------------|-------------------|
| INSPECTION TASK                                                                                                                                                                               | INSPECTION FREQUENCY |          | REFERENCED STANDARD | IBC REFERENCE | RESPONSIBLE AGENT |
|                                                                                                                                                                                               | CONTINUOUS           | PERIODIC |                     |               |                   |
| 1. INSPECTION OF COLD FORMED METAL ASSEMBLIES PER APPROVED...                                                                                                                                 |                      |          |                     |               |                   |
| A. MEMBER SIZE AND CONFIGURATION.                                                                                                                                                             | -                    | X        |                     |               |                   |
| B. CONNECTION SCREWS AND WELDING.                                                                                                                                                             | -                    | X        | AISI-S100           | 2211          | SI / SE           |
| C. MEMBER AND ASSEMBLY BRACING.                                                                                                                                                               | -                    | X        |                     |               |                   |
| 2. INSTALLATION PER APPROVED DETAILS:                                                                                                                                                         |                      |          |                     |               |                   |
| A. MEMBER SIZE AND SPACING.                                                                                                                                                                   | -                    | X        | AISI-S100           |               |                   |
| B. CONNECTION SCREWS.                                                                                                                                                                         | -                    | X        | AISI-S100           | 2211          | SI / SE           |
| C. WELDING.                                                                                                                                                                                   | -                    | X        | AISI-S100, AWS D1.3 |               |                   |
| D. BRIDGING/BRACING.                                                                                                                                                                          | -                    | X        | AISI-S100           |               |                   |
| 3. COLD-FORMED STEEL TRUSSES SPANNING 60 FEET OR GREATER:                                                                                                                                     |                      |          |                     |               |                   |
| A. VERIFY THAT THE TEMPORARY INSTALLATION RESTRAINT/BRAING AND THE PERMANENT INDIVIDUAL TRUSS MEMBER RESTRAINT/BRACING ARE INSTALLED IN ACCORDANCE WITH THE APPROVED TRUSS SUBMITTAL PACKAGE. | -                    | X        | AISI-S100           | 1705.2.4      | SI / SE           |
|                                                                                                                                                                                               |                      |          |                     |               |                   |



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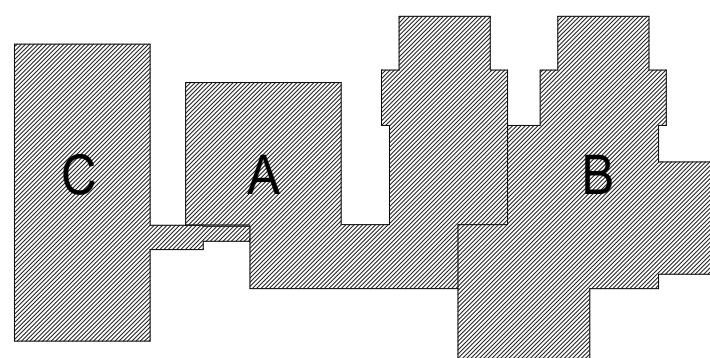
PROJECT TITLE

Starkweather  
Academy - Addition  
& Remodeling

Plymouth Canton  
Community Schools

DRAWING TITLE

SPECIAL INSPECTION  
SCHEDULES (CONT.)



Key Plan: NTS

ISSUE DATES

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| 01-30-2023 | CONSTRUCTION DOCUMENTS |
| 12-14-2022 | OWNER REVIEW           |
| 10-07-2022 | DESIGN DEVELOPMENT     |

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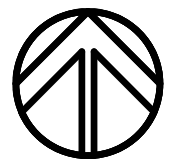
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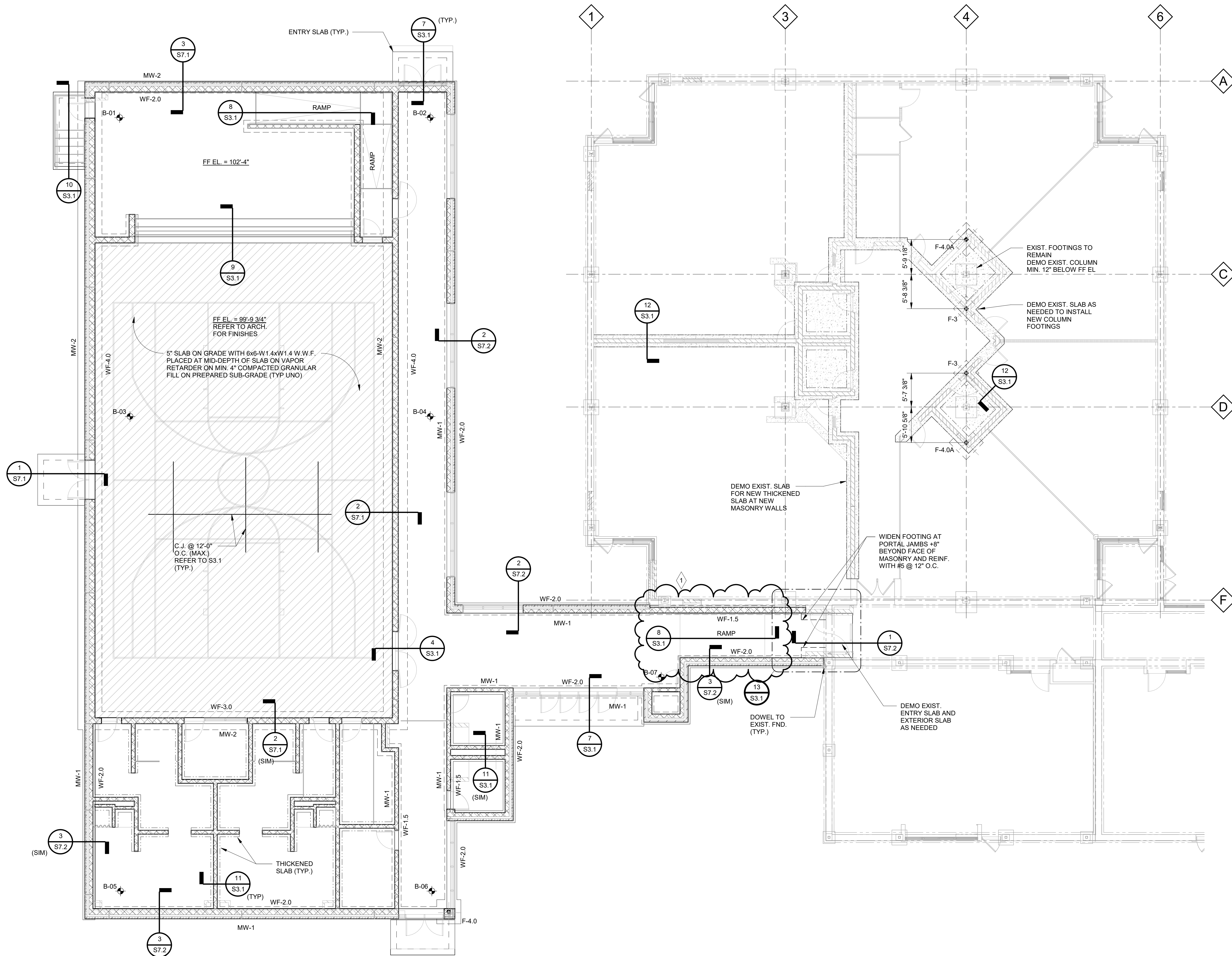


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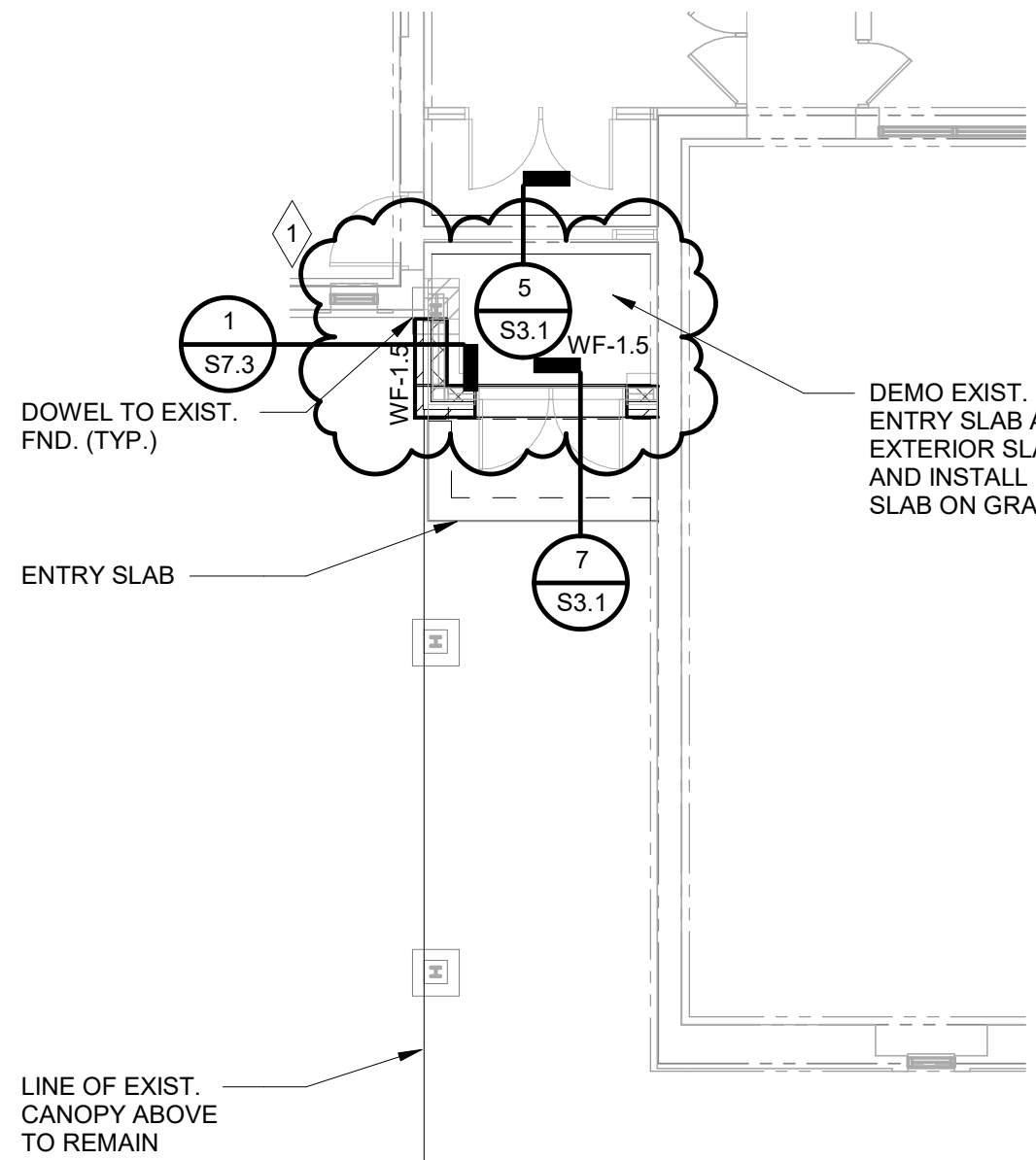
FOUNDATION PLAN - ZONE C

1/8" = 1'-0"



FOUNDATION PLAN - ZONE B

1/8" = 1'-0"



FOUNDATION NOTES:

- REFERENCE FINISHED FLOOR ELEVATION = 100'-0" BASED ON U.S.G.S. GEODETIC ELEVATION = 690.16'
- TOP OF FOOTING ELEVATION = 98'-8" UNLESS NOTED THUS [XX'-XX"]
- FOOTING ARE DESIGNED TO BEAR ON UNDISTURBED NATURAL SOILS WITH AN ASSUMED MINIMUM NET ALLOWABLE BEARING CAPACITY OF 1,500 PSF. DO NOT BEAR ON EXISTING FILL SOILS; EXTEND FOOTINGS AS NEED TO BEAR ON SUITABLE SOILS.
- CONTRACTOR SHALL COORDINATE ALL MASONRY DOWEL SIZES AND SPACING TO BE CAST INTO CONCRETE WITH MASONRY REINFORCING SHOP DRAWINGS.
- REFER TO CIVIL/SITE DRAWINGS FOR PROPOSED GRADE ELEVATIONS AROUND THE PERIMETER OF THE BUILDING.
- REFER TO MEP DRAWINGS FOR ALL PIPE AND CONDUIT SIZES AND LOCATIONS PASSING THROUGH AND/OR UNDER FOUNDATIONS.

DESIGNATIONS:

B-xx-xx APPROXIMATE LOCATION OF SOIL BORINGS. REFER TO GEOTECHNICAL REPORT NO. 225534 PREPARED BY G2 CONSULTING GROUP (DATED 10-27-2022)

F-3.0 3'-0"x3'-0"x12" DEEP (MIN.) FOOTING WITH (4) #4 BOTT. E.W.  
F-4.0 4'-0"x4'-0"x32" DEEP (MIN.) FOOTING WITH (4) #4 T&B. E.W.  
F-4.0A 4'-0"x4'-0"x12" DEEP (MIN.) FOOTING WITH (4) #4 BOTT. E.W.

WF-1.5: 18" WIDE x 32" DEEP (MIN.) WALL FOOTING WITH (2) #5 T&B  
WF-2.0: 24" WIDE x 32" DEEP (MIN.) WALL FOOTING WITH (2) #5 T&B  
WF-3.0: 36" WIDE x 32" DEEP (MIN.) WALL FOOTING WITH (3) #5 T&B  
WF-4.0: 48" WIDE x 32" DEEP (MIN.) WALL FOOTING WITH (4) #5 T&B

MW-1: 8" MASONRY WALL REINF. WITH #5 @ 48" O.C. VERT., UNO  
MW-2: 12" MASONRY WALL REINF. WITH #5 @ 48" O.C. VERT.

REFERENCE DRAWINGS:

S0.1-S0.2 GENERAL STRUCTURAL NOTES  
S0.3-S0.4 SPECIAL INSPECTION SCHEDULES  
S3.1 TYPICAL CONCRETE DETAILS  
S4.1 TYPICAL MASONRY DETAILS  
S5.1 TYPICAL STEEL DETAILS



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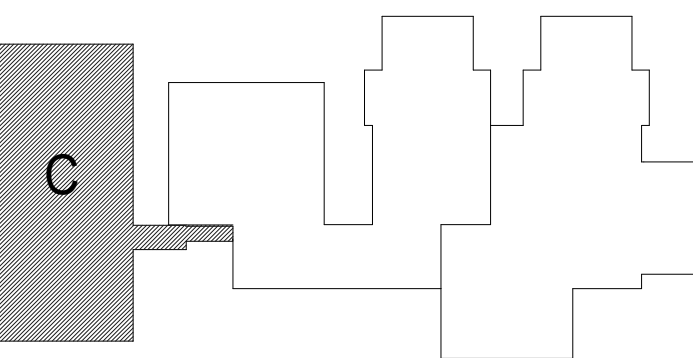
PROJECT TITLE

Starkweather Academy - Addition & Remodeling

Plymouth Canton Community Schools

DRAWING TITLE

FOUNDATION PLAN - ZONE C



Key Plan: NTS

ISSUE DATES

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| 02-20-2023 | ADDENDUM No 1          |
| 01-30-2023 | CONSTRUCTION DOCUMENTS |
| 12-14-2022 | OWNER REVIEW           |
| 10-07-2022 | DESIGN DEVELOPMENT     |

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| DRAWN    | DD          |
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| APPROVED | ANN         |

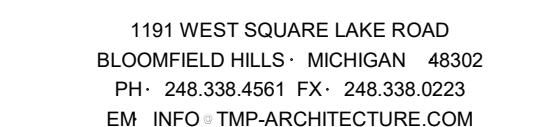
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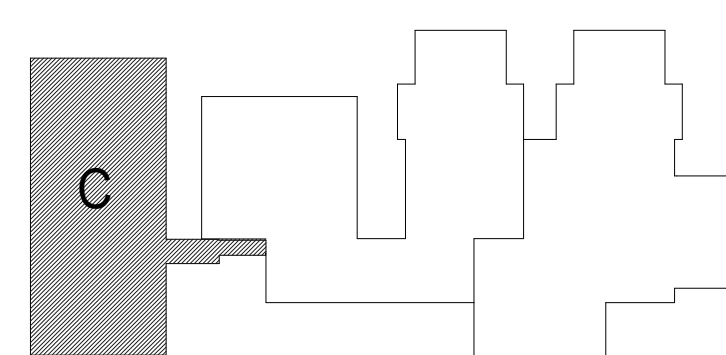
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PROJECT TITLE

# Starkweather Academy - Addition & Remodeling

Plymouth Canton  
Community Schools

DRAWING TITLE  
ROOF FRAMING PLAN -  
ZONE C



 Key Plan: NTS

## ISSUE DATES

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| 01-30-2023 | CONSTRUCTION DOCUMENTS |
| 12-14-2022 | OWNER REVIEW           |
| 10-07-2022 | DESIGN DEVELOPMENT     |

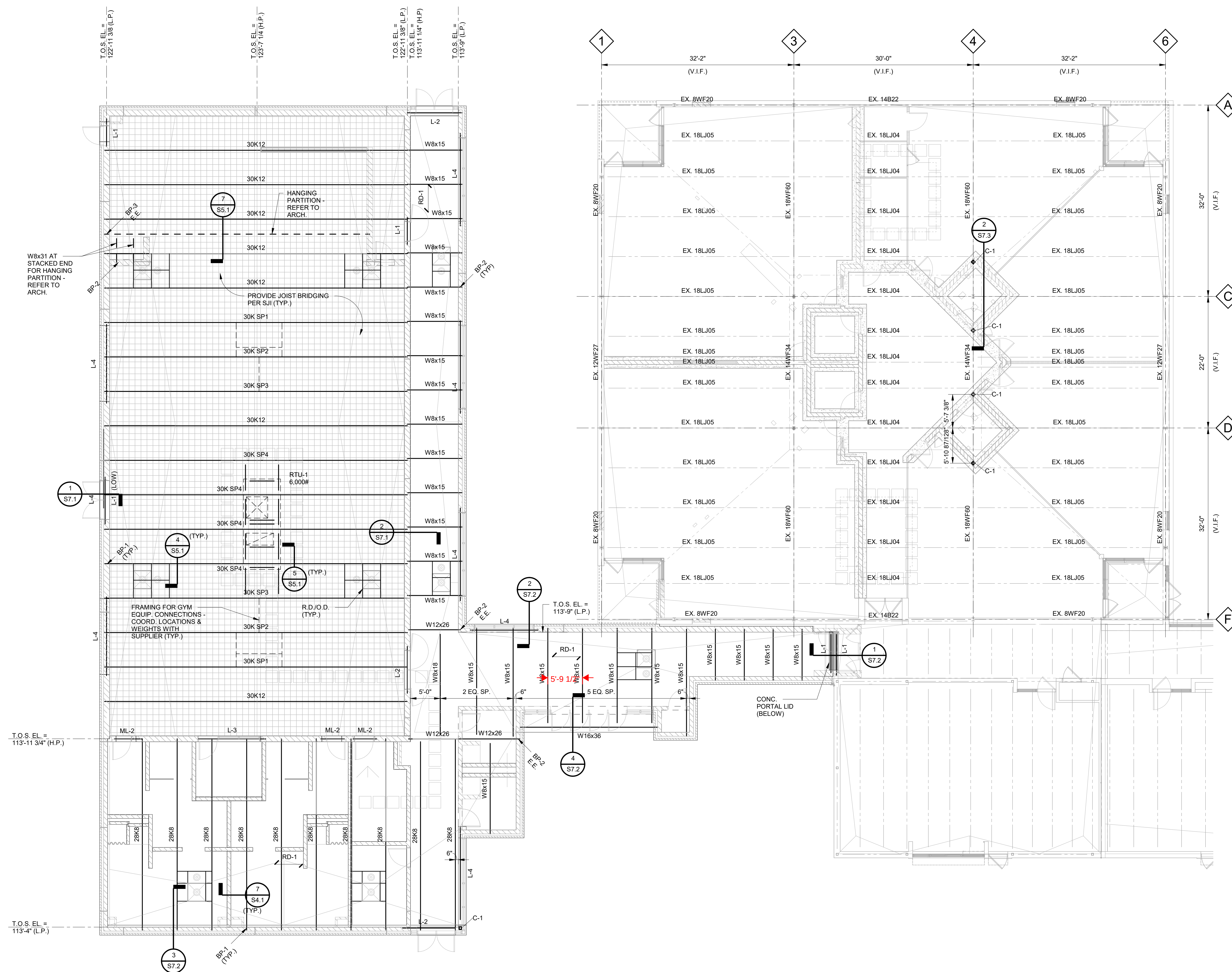
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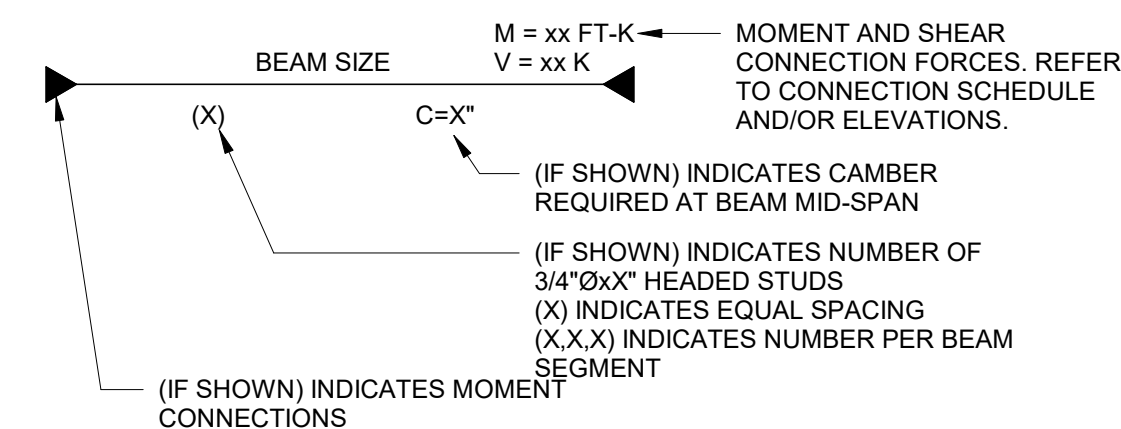
DRAWING NO.

## S1.2C



ROOF FRAMING NOTES:

1. TOP OF SLAB REFERENCE. ELEVATION (DECK BEARING ELEVATION) = REFER TO PLAN  
☐ INDICATES HIGH ROOF AREA
2. DESIGNATIONS:
- RD-1: 1.5'-20" GAGE MIN. TYPE "32" WIDE RIB GALVANIZED STEEL ROOF DECK (MIN. 3 SPAN CONT.)  
 REFER TO DRAWING SS-1 FOR ATTACHMENT DETAILS  
 MINIMUM DECK SECTION PROPERTIES FOR DECK BASED ON FY = 40 KSI:  
 DESIGN THICKNESS = 0.0358" (UNCAOATED)  
 I (POSITIVE) = 0.201 IN<sup>4</sup>/FT  
 I (NEGATIVE) = 0.222 IN<sup>4</sup>/FT  
 S (POSITIVE) = 0.234 IN<sup>3</sup>/FT  
 I (NEGATIVE) = 0.247 IN<sup>4</sup>/FT
- C-1: HSS5x5/16 WITH 12"x12"x3/4" BASE PLATE AND (4) 3/4" Ø ANCHOR RODS
- BP-X BEARING PLATE, REFER TO SCHEDULE
- L-X LINTEL, REFER TO SCHEDULE
4. ALL JOIST SEATS FOR K-SERIES JOISTS SHALL BE 2 1/2" DEEP UNLESS NOTED OTHERWISE.
4. ALL JOISTS SHALL BE DESIGNED FOR A NET UPLIFT OF 26 PSF (LRFD), IN ADDITION TO OTHER LOADS AND ALL OTHER NON-UNIFORM LOADS INDICATED ON THE DRAWINGS. ALL BRIDGING AND UPLIFT BRIDGING SHALL BE PER SJ REQUIREMENTS.  
 PROVIDE 1/4" CAP PLATE ON COLUMN FOR DECK BEARING AS REQUIRED.
5. PROVIDE 1/4" CAP PLATE ON COLUMN FOR DECK BEARING AS REQUIRED.
6. COORDINATE SIZES AND LOCATION OF ALL ROOF OPENINGS WITH ARCHITECTURAL AND MEP DRAWINGS. VERIFY SIZES, WEIGHTS, AND LOCATIONS OF ROOF TOP MECHANICAL EQUIPMENT WITH ARCHITECTURAL AND MEP DRAWINGS. REPORT ANY DISCREPANCIES WITH RESPECT TO SIZE AND WEIGHTS SHOWN ON THESE DRAWINGS TO THE ARCHITECT AND ENGINEER.
7. FRAMING FOR ALL ROOF DRAINS AND OVERFLOW DRAINS SHALL BE L5x3 1/2x5/16 LLV TYPICAL. ALL SIDES OF SUPPORTED EDGE, UNLESS NOTED OTHERWISE.
8. BEAM DESIGNATIONS:



9. REFERENCE DRAWINGS:
- |           |                              |
|-----------|------------------------------|
| S0.1-S0.2 | GENERAL STRUCTURAL NOTES     |
| S0.3-S0.4 | SPECIAL INSPECTION SCHEDULES |
| S3.1      | TYPICAL CONCRETE DETAILS     |
| S4.1      | TYPICAL MASONRY DETAIL       |
| S5.1      | TYPICAL STEEL DETAILS        |

ROOF FRAMING PLAN - ZONE C

$$1/8'' = 1'-0''$$





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PROJECT TITLE

# Starkweather Academy - Addition & Remodeling

Plymouth Canton  
Community Schools

DRAWING TITLE  
TYPICAL CONCRETE  
DETAILS



## ISSUE DATES

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| 01-30-2023 | CONSTRUCTION DOCUMENTS |
| 12-14-2022 | OWNER REVIEW           |
| 10-07-2022 | DESIGN DEVELOPMENT     |

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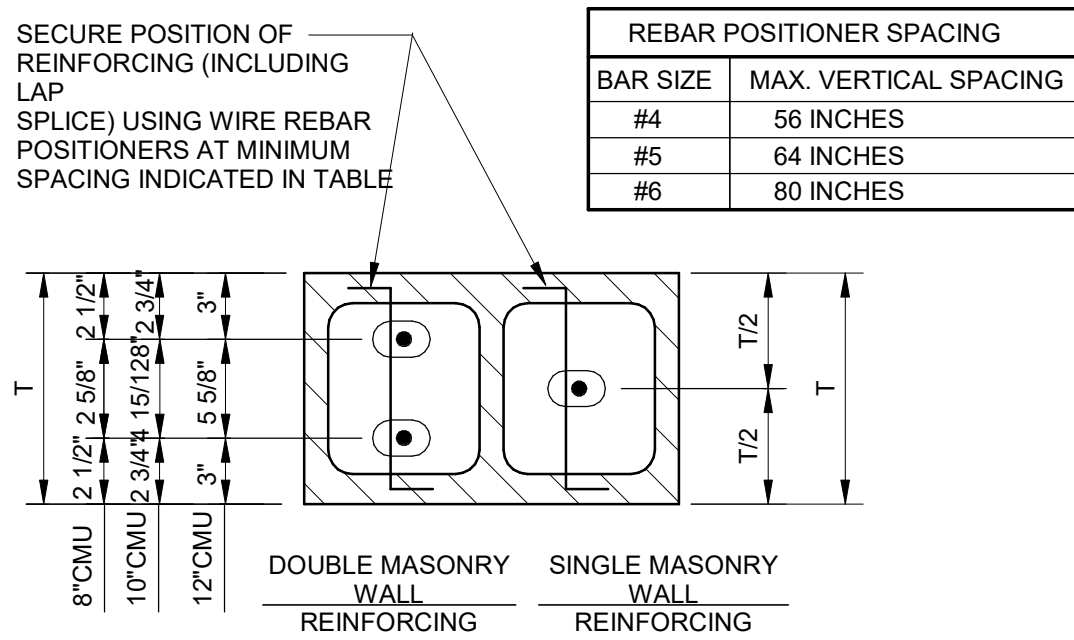
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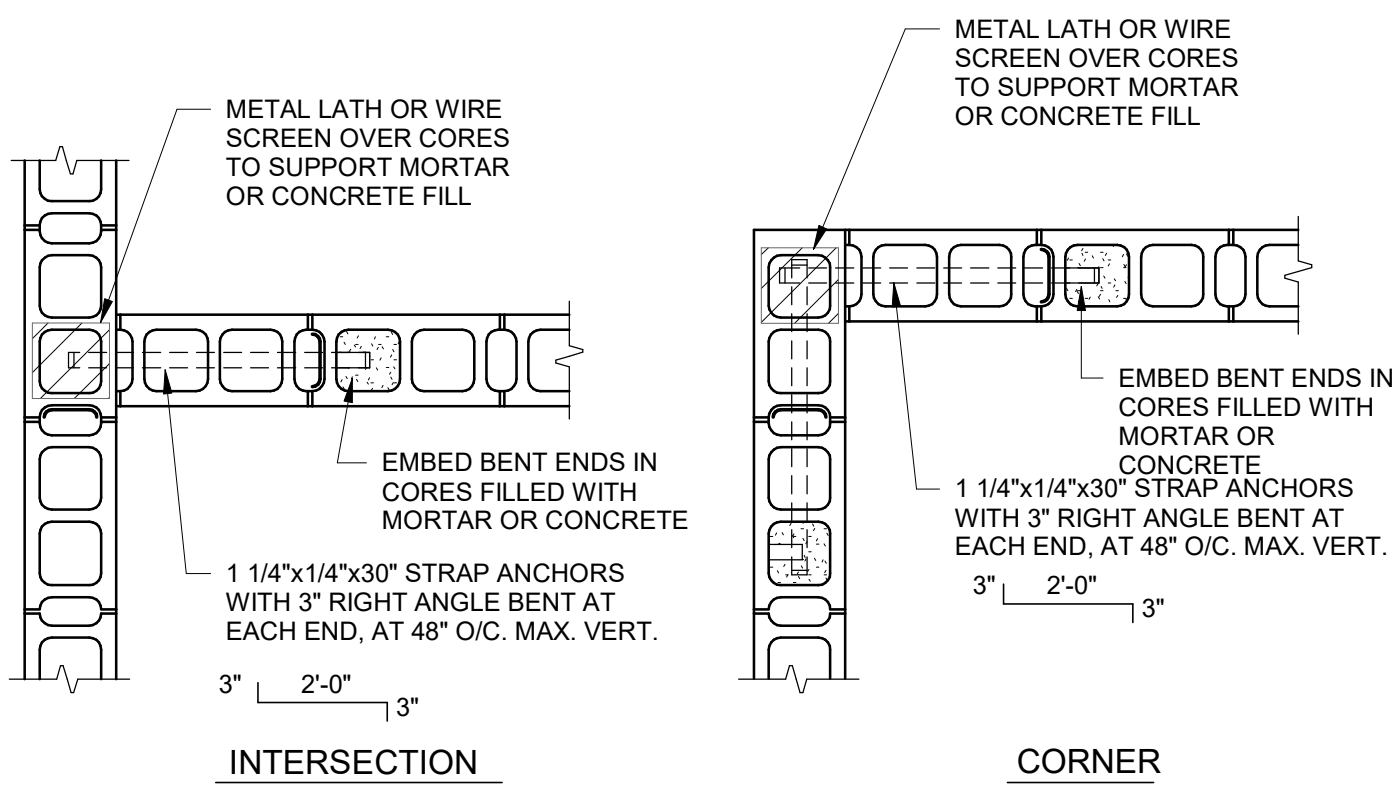
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### S3.1

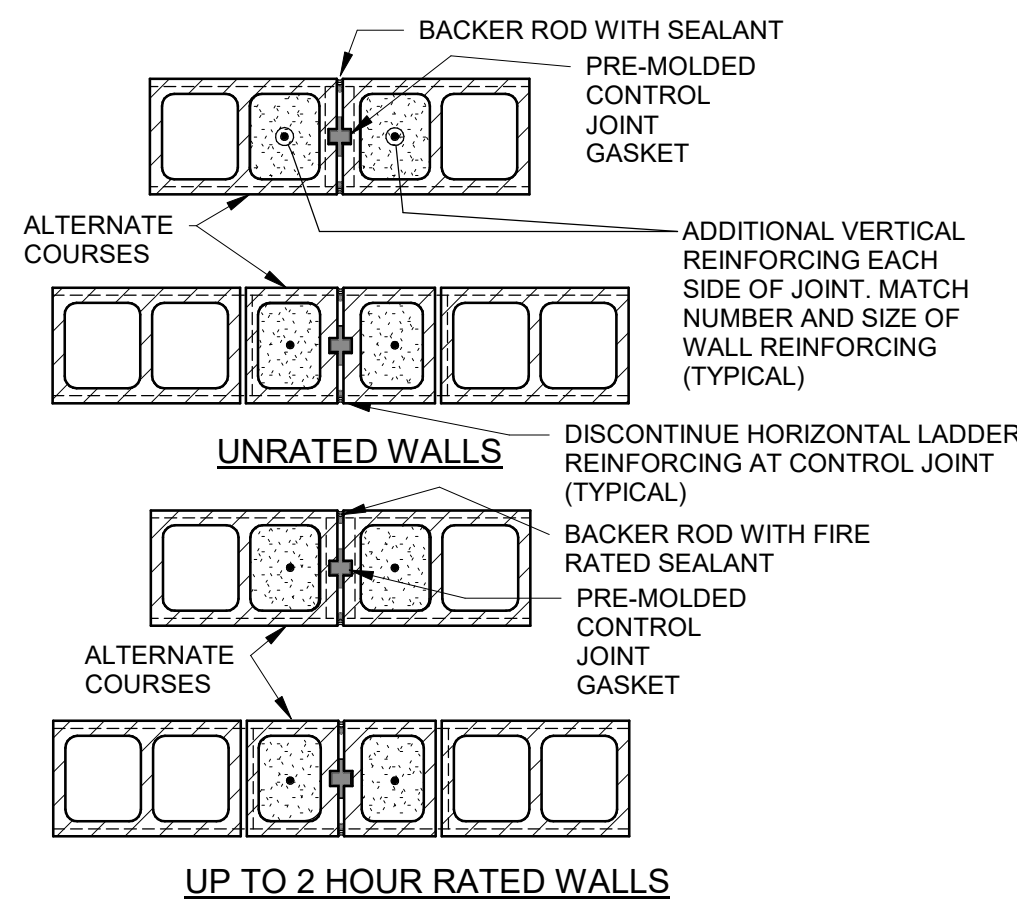




1 TYP. MASONRY WALL REINF.  
S4.1 1 1/2" = 1'-0"



2 TYPICAL PLANS AT MASONRY WALL INTERSECTION & CORNER  
S4.1 3/4" = 1'-0"



3 CONTROL JOINT DETAILS  
S4.1 3/4" = 1'-0"

| STEEL LINTEL SCHEDULE |                   |               |         |
|-----------------------|-------------------|---------------|---------|
| MARK                  | SIZE              | BEARING PLATE | REMARKS |
| L-1                   | W8x18 + 3/8" PL.  | BP-2 E.E.     | .       |
| L-2                   | W8x24 + 3/8" PL.  | BP-2 E.E.     |         |
| L-3                   | W16x26 + 3/8" PL. | BP-2 E.E.     |         |
| L-4                   | W16x36 + 3/8" PL. | BP-2 E.E.     |         |
|                       |                   |               |         |
|                       |                   |               |         |

| MASONRY LINTEL SCHEDULE |                |               |         |
|-------------------------|----------------|---------------|---------|
| MARK                    | SIZE           | REINFORCING   | REMARKS |
| ML-1                    | 8" CMU x 8" D  | (2) #5 BOTTOM |         |
| ML-2                    | 12" CMU x 8" D | (2) #5 BOTTOM |         |
|                         |                |               |         |
|                         |                |               |         |

- NOTES
- PLACE LINTEL BEAMS CENTERED IN WALLS UNLESS NOTED OTHERWISE.
  - LINTELS NOT SPECIFICALLY SIZED BY DRAWINGS AND DETAIL SECTIONS SHALL COMPLY WITH THE FOLLOWING:
    - VENEERS OF 4" OR LESS IN THICKNESS:
      - UP TO 4'-0" WIDE OPENING: L3x3 1/2x5/16 LLV
      - OVER 4'-0" UP TO 6'-0" WIDE OPENING: L5x3 1/2x5/16 LLV
      - OVER 6'-0" UP TO 8'-0" WIDE OPENING: L7x3 1/2x5/16 LLV
    - WALLS 8" OR MORE IN THICKNESS (FOR EA. 4" WIDTH OF MASONRY THICKNESS): STITCH WELD DOUBLE ANGLES TOGETHER @ 12" O.C. MAX. TOP & BOTTOM.
      - UP TO 4'-0" WIDE OPENING: L 3 1/2 x 4 x 5/16 LLV OR 8" D CMU LINTEL WITH (2) #4 CONT. BOTTOM.
      - OVER 4'-0" UP TO 7'-0" WIDE OPENING: L 3 1/2 x 5 x 5/16 LLV OR 8" D CMU LINTEL WITH (2) #5 CONT. BOTTOM.
      - OVER 7'-0" WIDE OPENING: STRUCTURAL LINTEL REQUIRED
    - ABOVE LINTELS TO HAVE 8" BEARING AT EACH END.

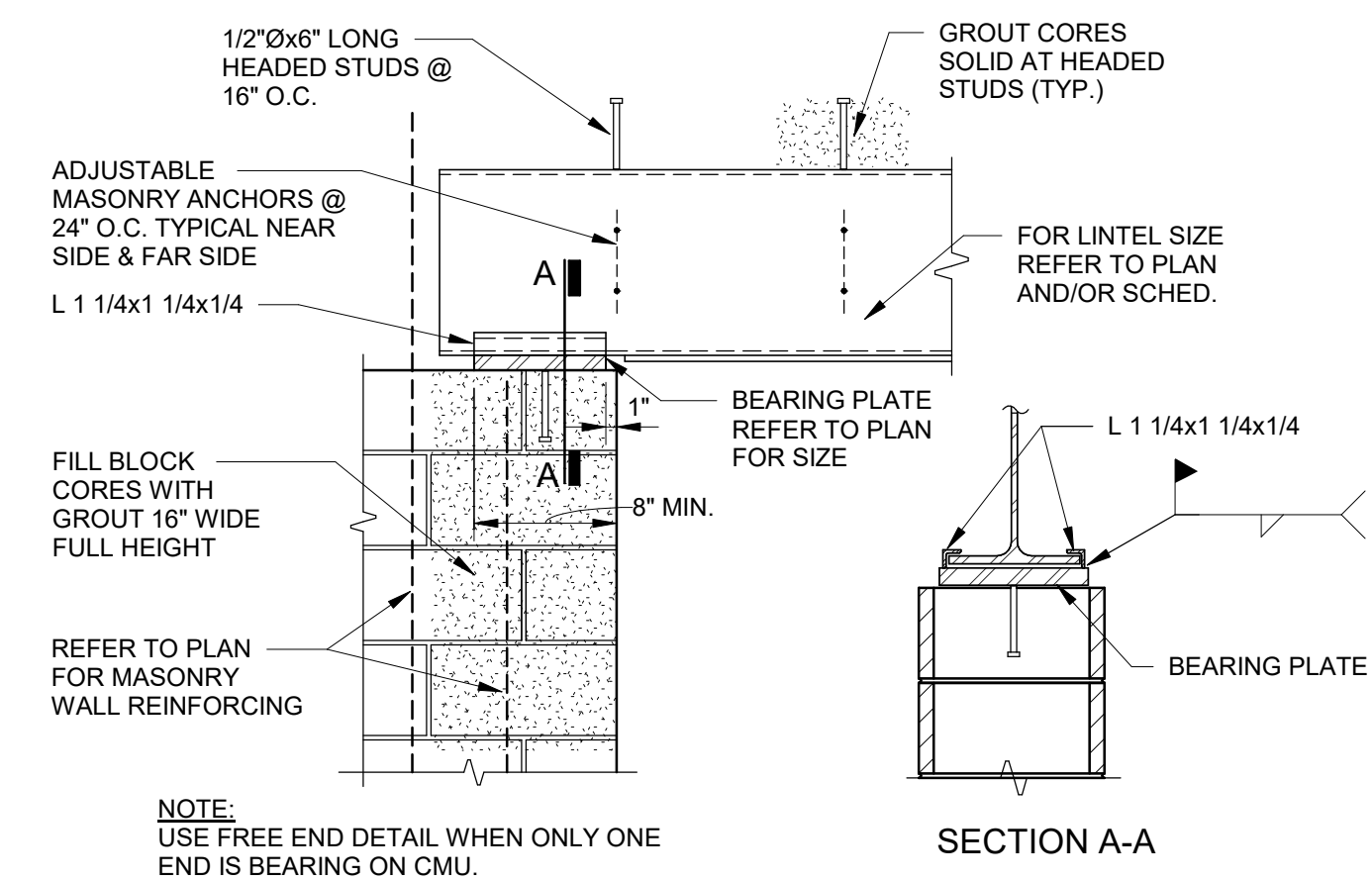
| BEARING PLATE SCHEDULE |                     |                         |         |
|------------------------|---------------------|-------------------------|---------|
| MARK                   | SIZE (L" x W" x T") | ANCHORAGE               | REMARKS |
| BP-1                   | 6"x6"x1/2"          | (2) 1/2" Ø HEADED STUDS |         |
| BP-2                   | 7"x7"x1/2"          | (2) 1/2" Ø HEADED STUDS |         |
| BP-3                   | 7"x11"x1/2"         | (2) 1/2" Ø HEADED STUDS |         |
|                        |                     |                         |         |

NOTE:  
EDGE OF BEARING PLATE SHALL BE LOCATED NOT MORE THAN 1/2" FROM THE FACE OF THE WALL ON THE SIDE OF BEAM OR JOIST SPAN.

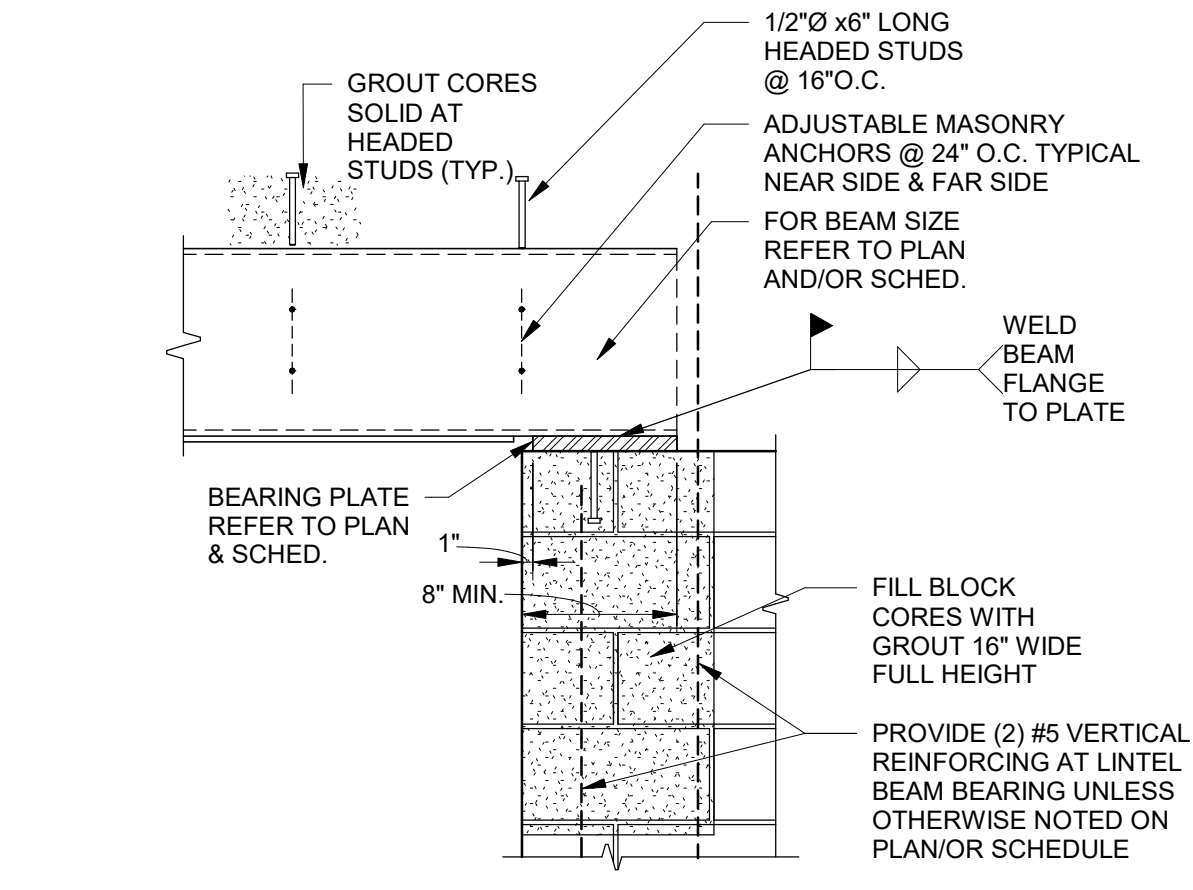
| MASONRY VERTICAL REINFORCING LAP SPLICE LENGTH SCHEDULE |                                |           |           |                                  |           |           |                                  |           |           |                                   |           |           |
|---------------------------------------------------------|--------------------------------|-----------|-----------|----------------------------------|-----------|-----------|----------------------------------|-----------|-----------|-----------------------------------|-----------|-----------|
|                                                         | 8" CMU WITH VERT BARS CENTERED |           |           | 8" CMU WITH VERT. BARS EACH FACE |           |           | 12" CMU WITH VERT. BARS CENTERED |           |           | 12" CMU WITH VERT. BARS EACH FACE |           |           |
| f <sub>m</sub> =                                        | 2,000 PSI                      | 2,500 PSI | 3,000 PSI | 2,000 PSI                        | 2,500 PSI | 3,000 PSI | 2,000 PSI                        | 2,500 PSI | 3,000 PSI | 2,000 PSI                         | 2,500 PSI | 3,000 PSI |
| #4                                                      | 14"                            | 12"       | 12"       | 20"                              | 18"       | 16"       | 12"                              | 12"       | 12"       | 16"                               | 15"       | 13"       |
| #5                                                      | 21"                            | 19"       | 18"       | 32"                              | 28"       | 26"       | 13"                              | 12"       | 12"       | 26"                               | 24"       | 21"       |
| #6                                                      | 31"                            | 28"       | 26"       | 47"                              | 42"       | 38"       | 19"                              | 17"       | 16"       | 38"                               | 34"       | 31"       |
| #7                                                      | 43"                            | 39"       | 35"       | 65"                              | 58"       | 53"       | 27"                              | 24"       | 22"       | 53"                               | 47"       | 43"       |
| #8                                                      | 57"                            | 51"       | 47"       | 88"                              | 78"       | 72"       | 35"                              | 31"       | 29"       | 70"                               | 63"       | 57"       |
| #9                                                      | 74"                            | 66"       | 61"       | 114"                             | 102"      | 94"       | 45"                              | 40"       | 37"       | 91"                               | 81"       | 74"       |
|                                                         |                                |           |           |                                  |           |           |                                  |           |           |                                   |           |           |

NOTES:

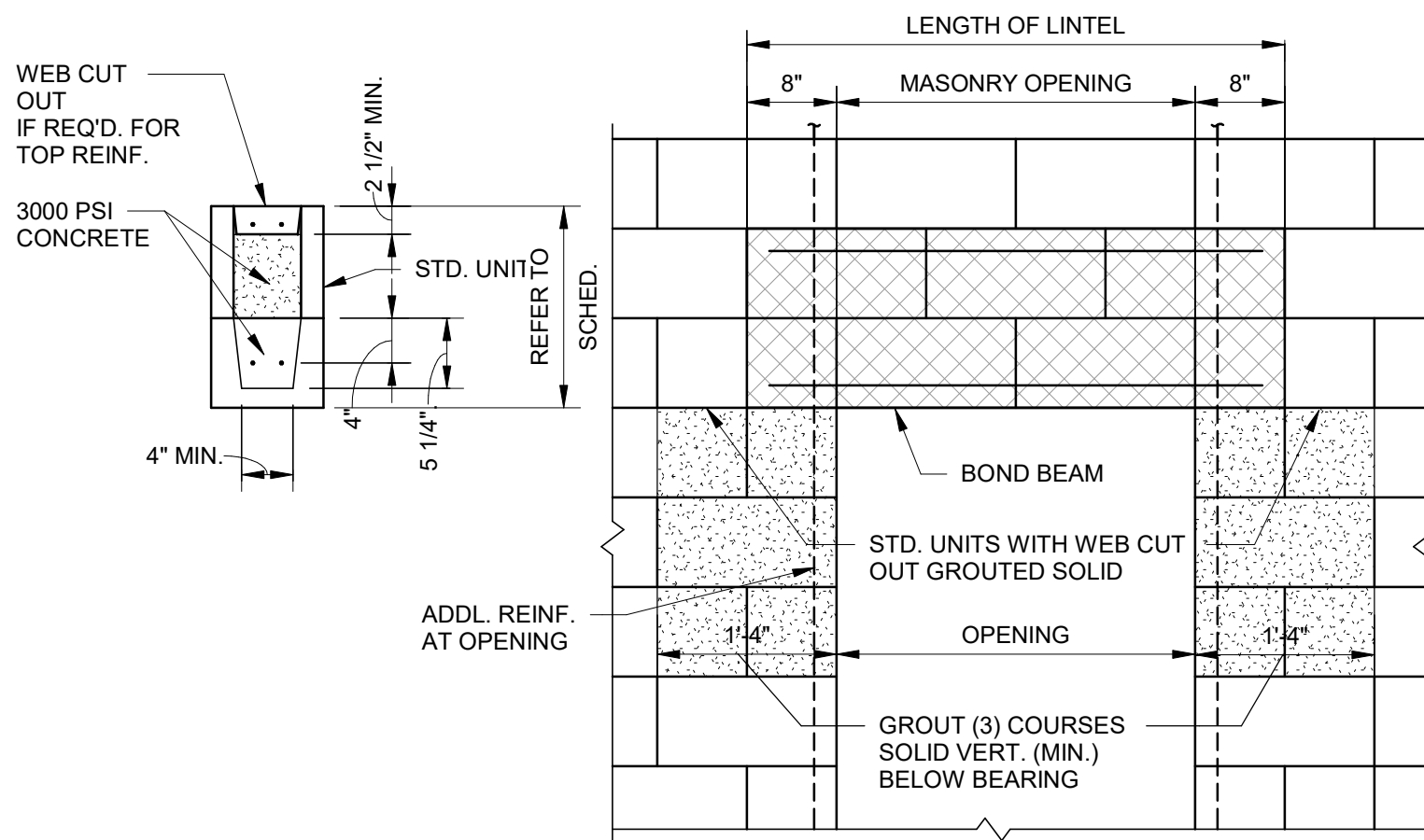
- VERTICAL BARS SHALL BE HELD IN POSITION BY WIRE TIES OR OTHER APPROVED MEANS TO INSURE DESIGN LOCATIONS AND LAP. REFER TO DETAIL 1/54.1.
- REFER TO PLAN FOR VERTICAL REINFORCING.



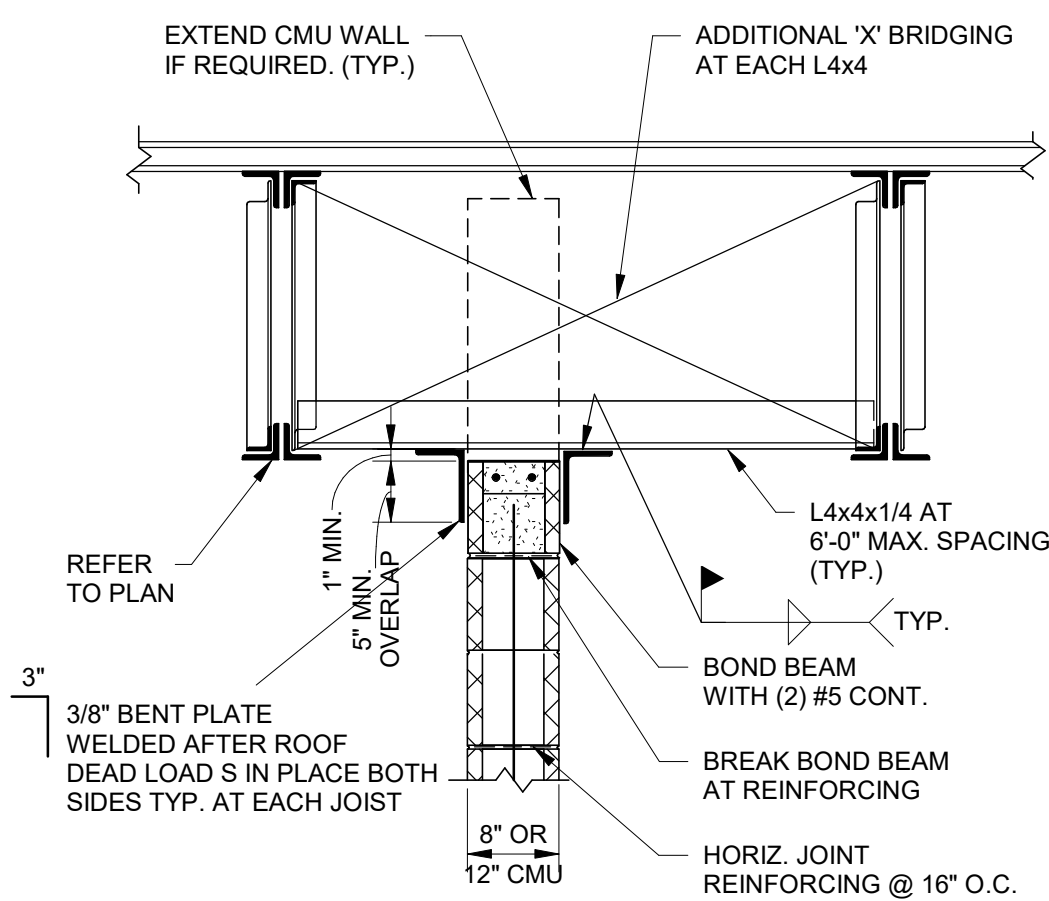
4 LINTEL BEARING DETAIL AT FREE END  
S4.1 3/4" = 1'-0"



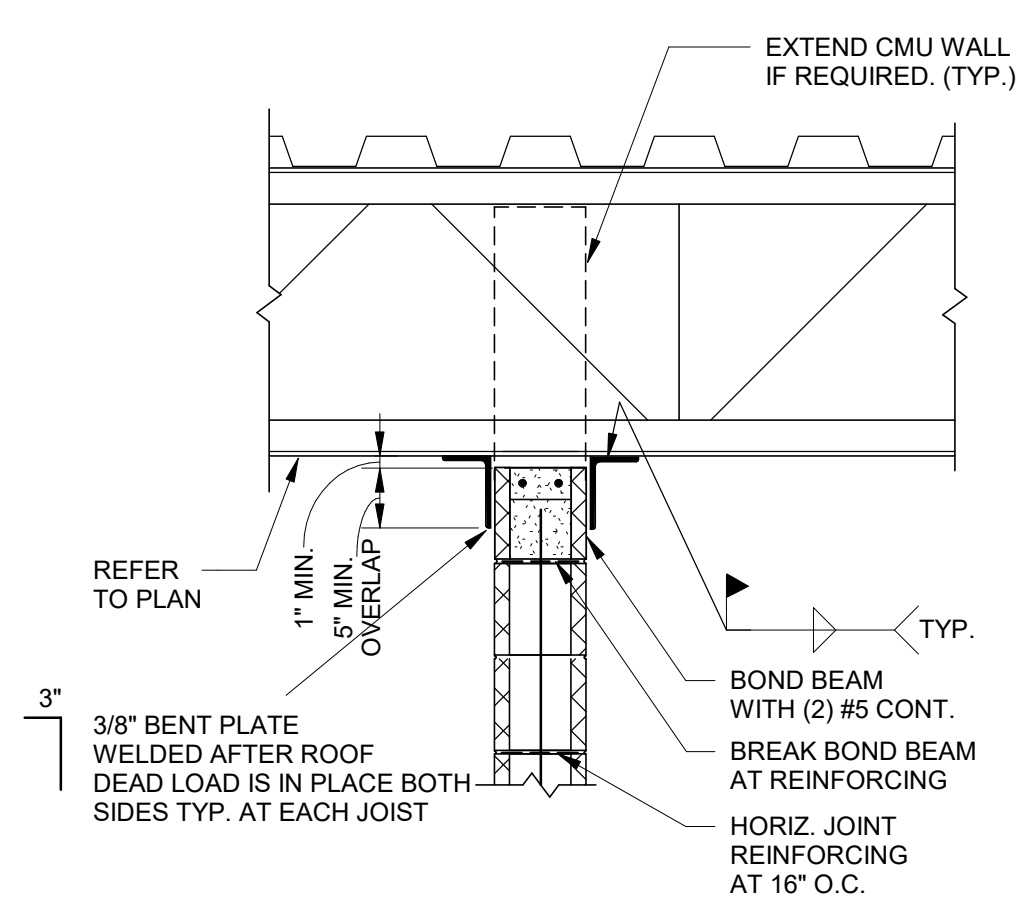
5 LINTEL BEARING DETAIL AT FIXED END  
S4.1 3/4" = 1'-0"



6 MASONRY LINTEL DETAIL  
S4.1 3/4" = 1'-0"



7 MASONRY WALL BRACING PARALLEL TO JOIST  
S4.1 3/4" = 1'-0"



8 MASONRY WALL BRACING PERPENDICULAR TO JOIST  
S4.1 3/4" = 1'-0"



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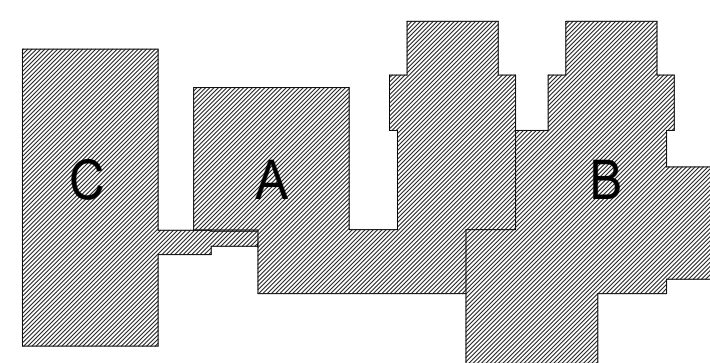
PROJECT TITLE

Starkweather  
Academy - Addition  
& Remodeling

Plymouth Canton  
Community Schools

DRAWING TITLE

TYPICAL MASONRY  
DETAILS



Key Plan: NTS

ISSUE DATES

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01-30-2023 CONSTRUCTION DOCUMENTS  
12-14-2022 OWNER REVIEW  
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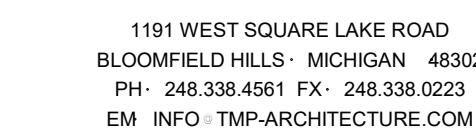
S4.1





**NOTES:**

1. PROVIDE JOIST TOP OR BOTTOM CHORD REINFORCING PER DETAIL 4/S5.1 WHERE LOADS ARE NOT LOCATED WITHIN 6" OF JOIST WEB MEMBERS.
2. LOADS INDICATED IN SCHEDULE ARE POSITIVE DOWNWARD AND NEGATIVE UPWARD. FINAL LOADS SHALL BE CONFIRMED WITH EQUIPMENT SUPPLIER(S).
3. PROVIDE ADDITIONAL FRAMING WHERE EQUIPMENT HANGERS ARE LOCATED BETWEEN JOISTS.
4. EQUIPMENT LOADS MAY BE HUNG FROM TOP OR BOTTOM CHORDS. COORDINATE CONNECTION REQUIREMENTS WITH EQUIPMENT SUPPLIER.
5. COORDINATE ALL DIMENSIONS WITH ARCH. AND EQUIPMENT SUPPLIER.



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PROJECT TITLE

# Starkweather Academy - Addition & Remodeling

Plymouth Canton  
Community Schools

DRAWING TITLE  
TYPICAL STEEL DETAILS



## ISSUE DATES

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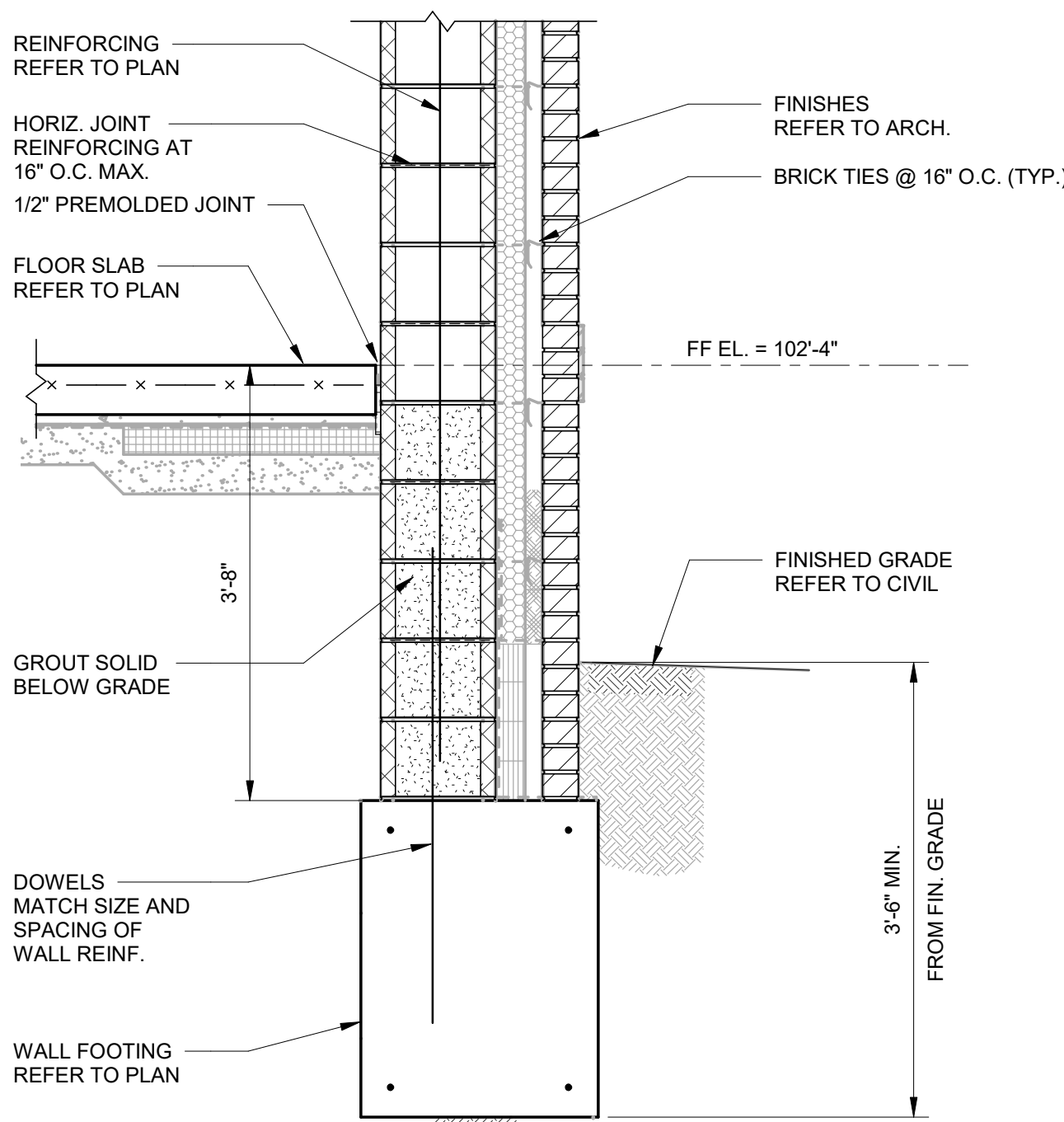
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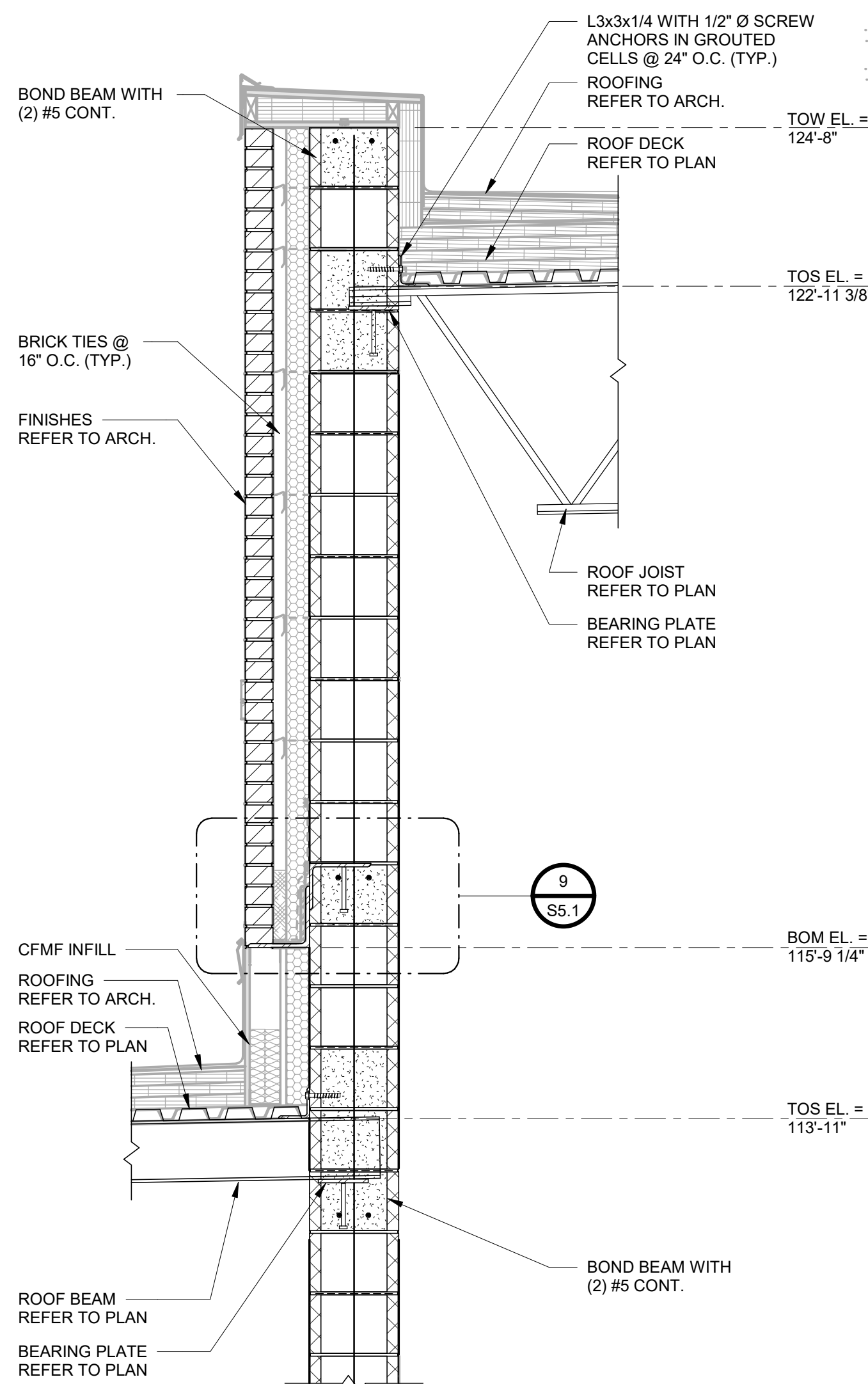
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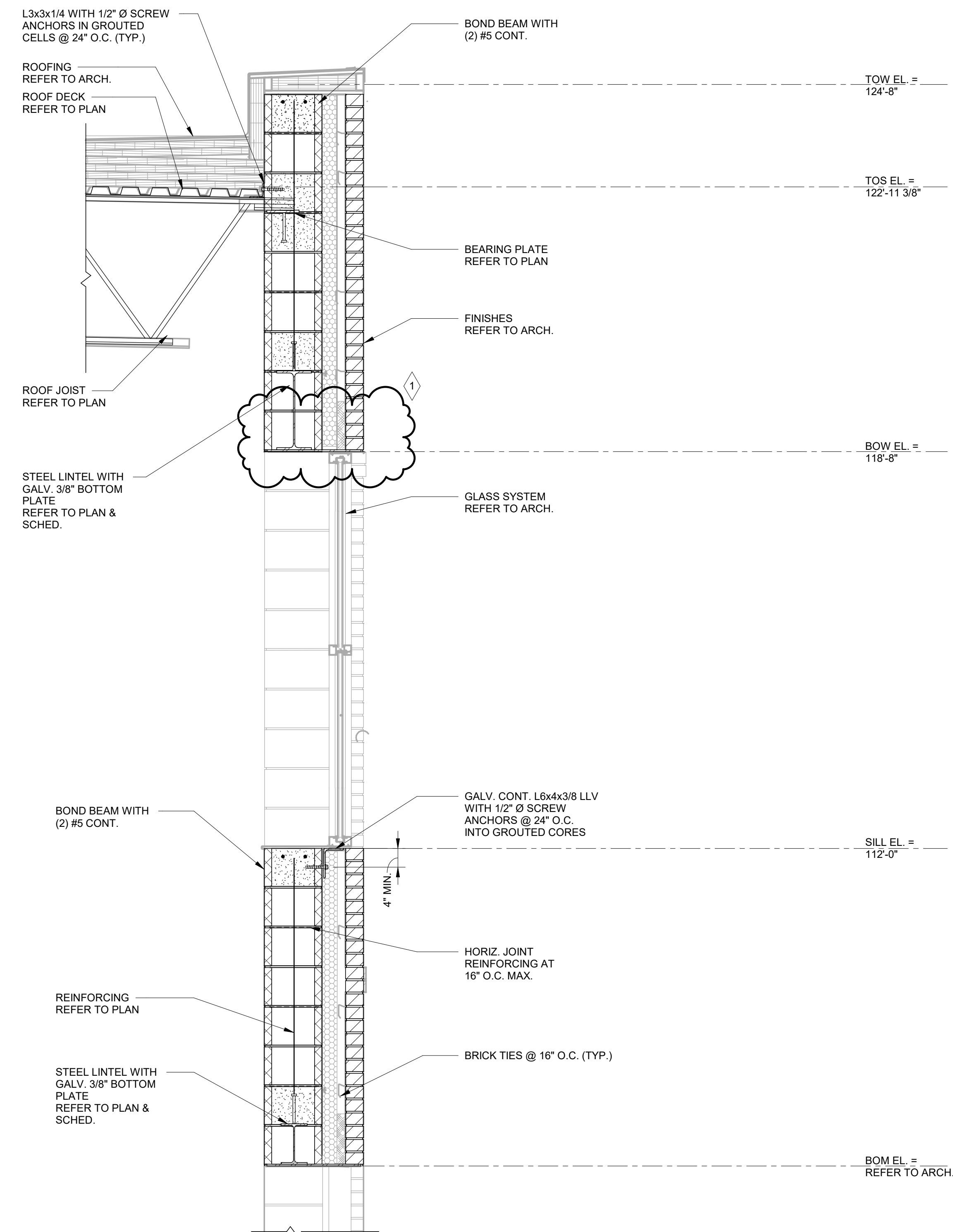
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2/15/2023 12:30:23 PM



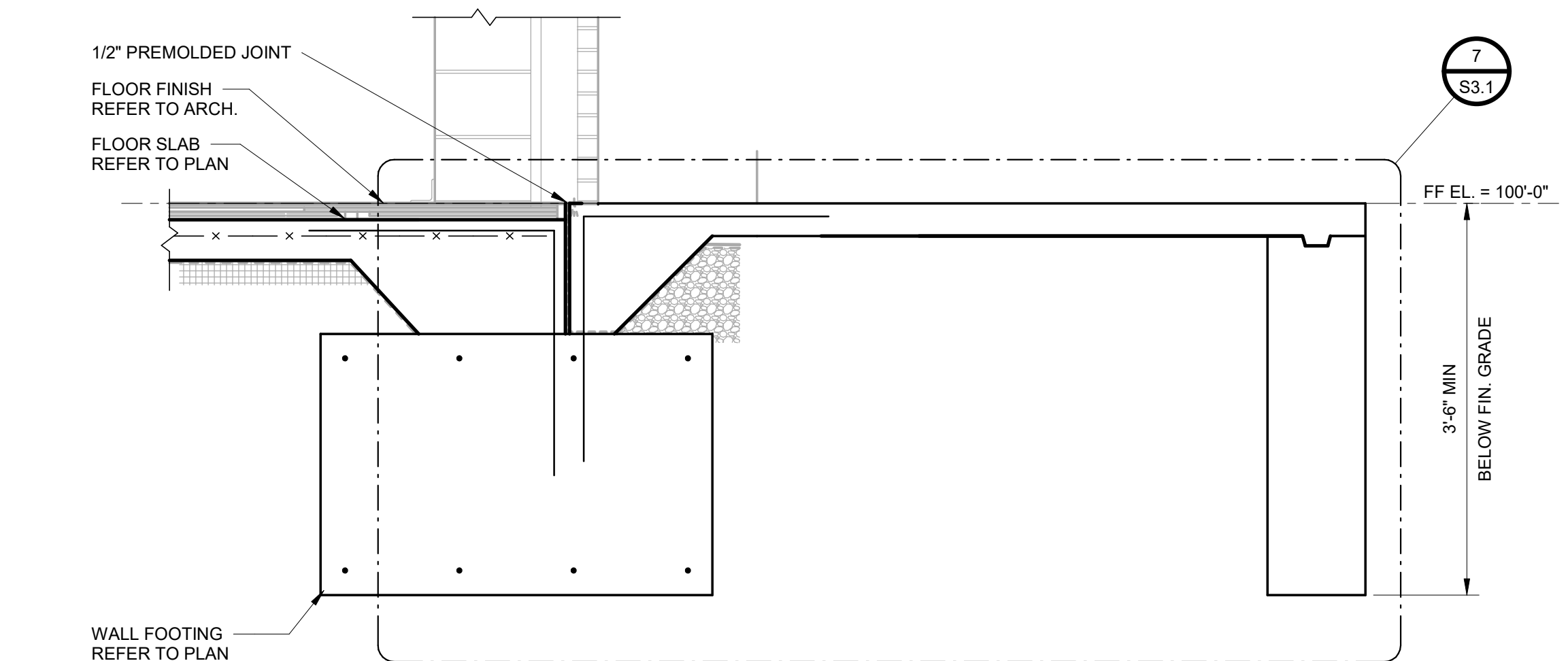
3 SECTION  
S7.1 3/4" = 1'-0"



2 SECTION  
S7.1 3/4" = 1'-0"



1 SECTION  
S7.1 3/4" = 1'-0"



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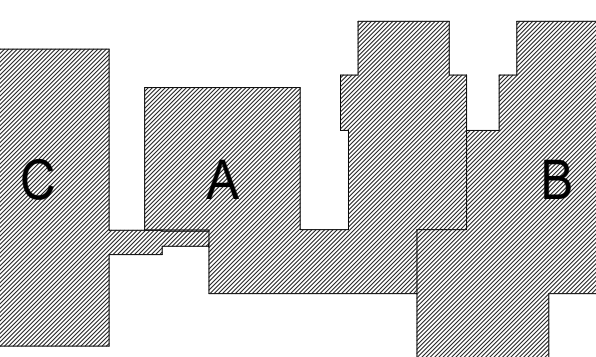
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Plymouth Canton Community Schools

DRAWING TITLE  
SECTIONS AND DETAILS



Key Plan: NTS

ISSUE DATES

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| 02-20-2023 | ADDENDUM No 1          |
| 01-30-2023 | CONSTRUCTION DOCUMENTS |
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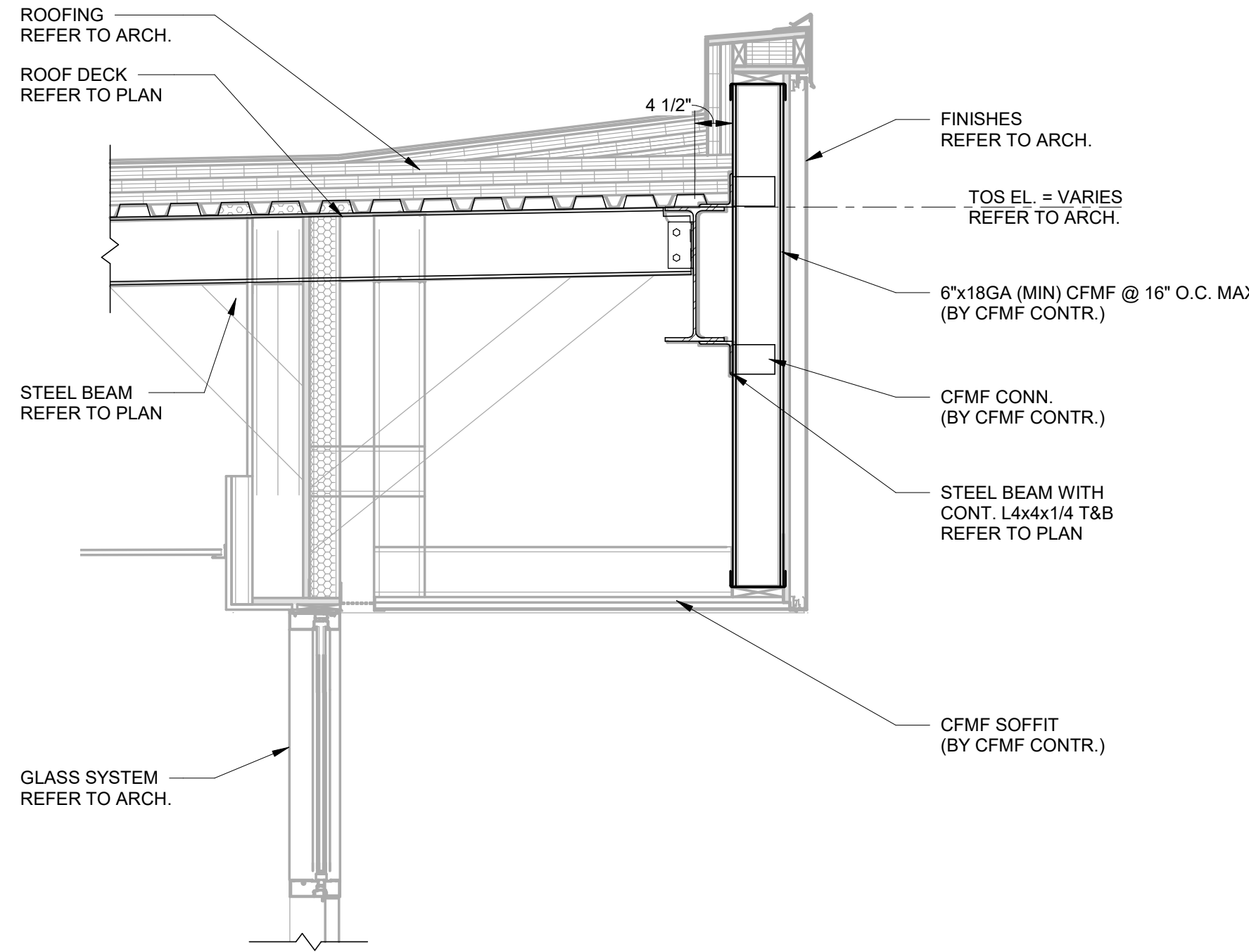
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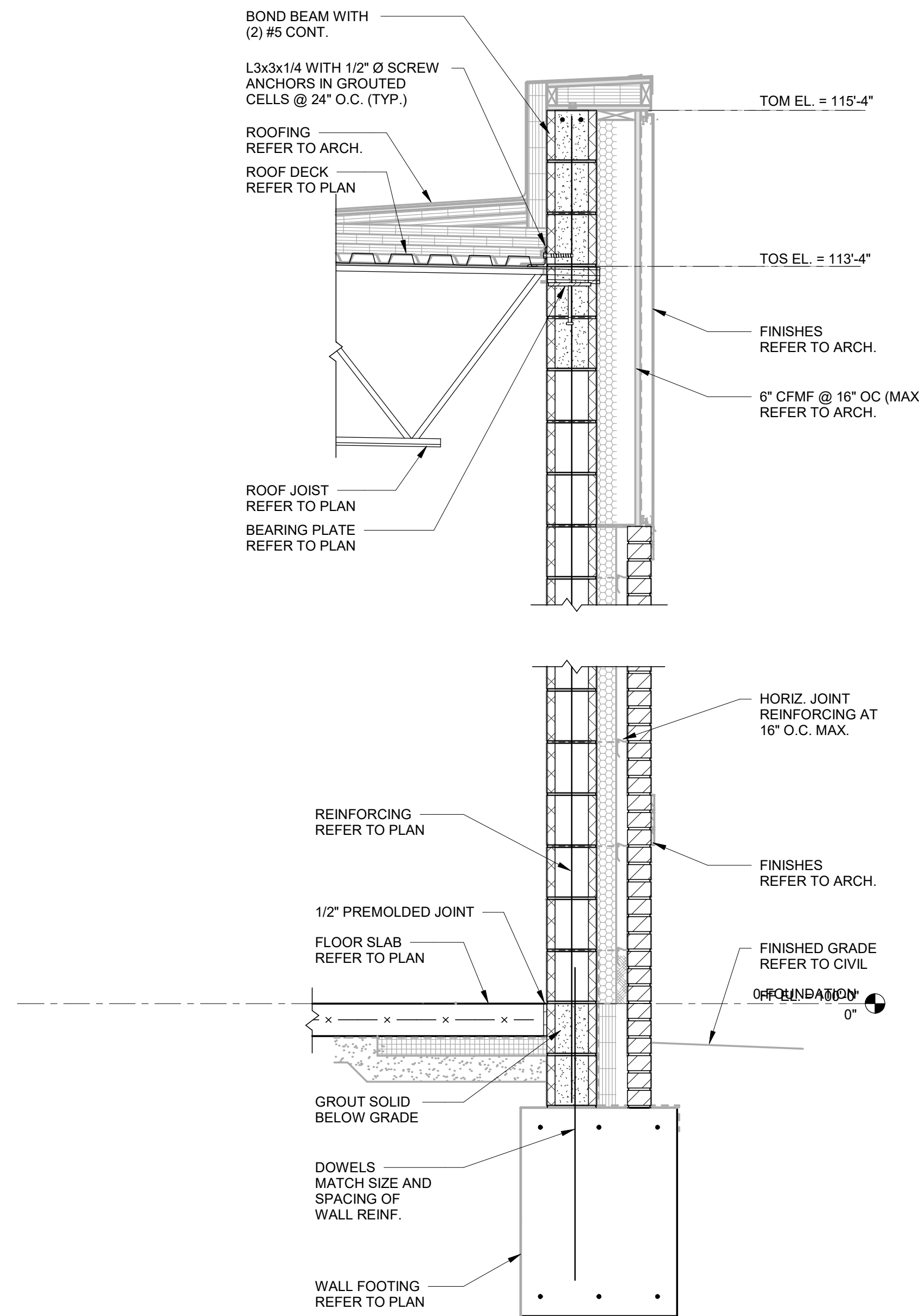
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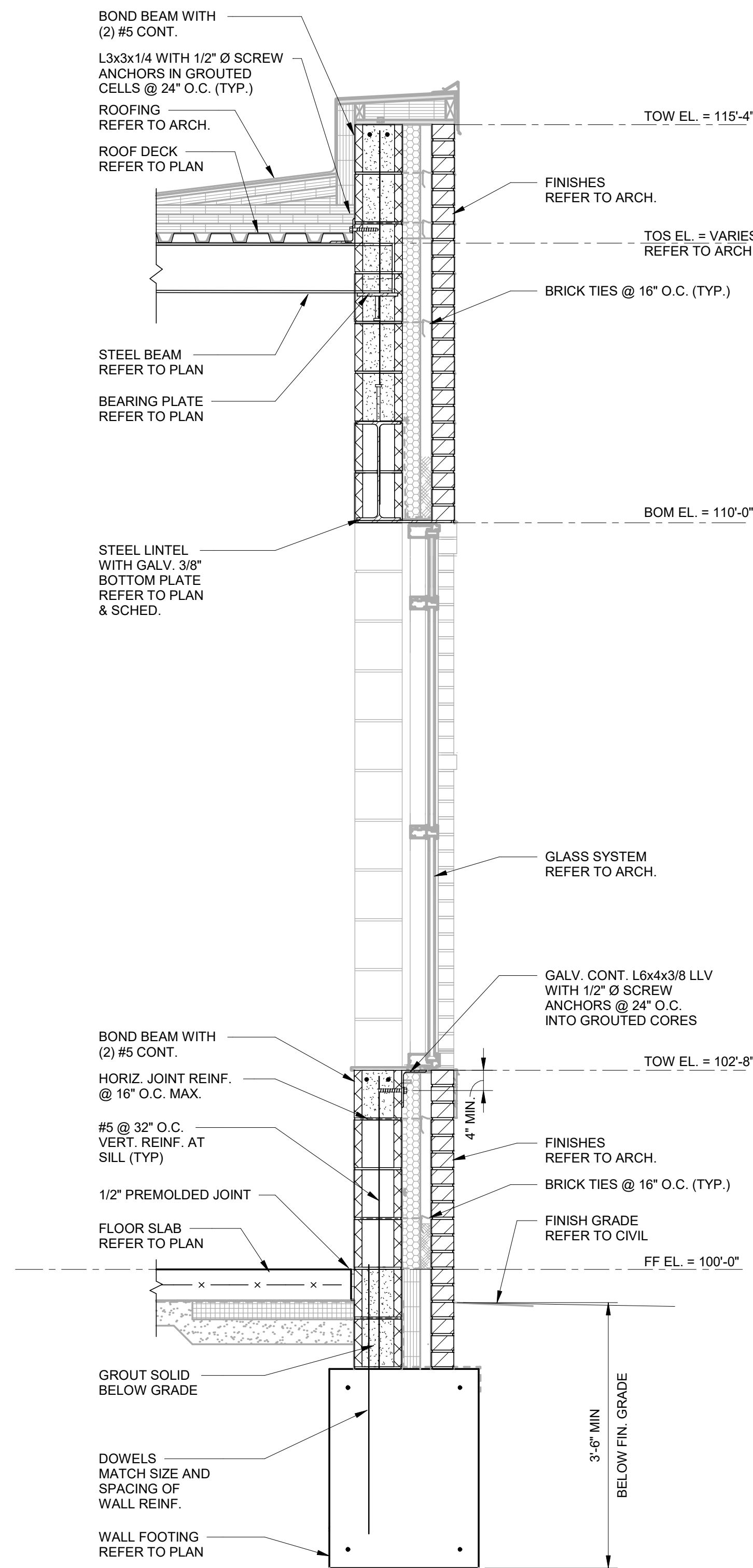




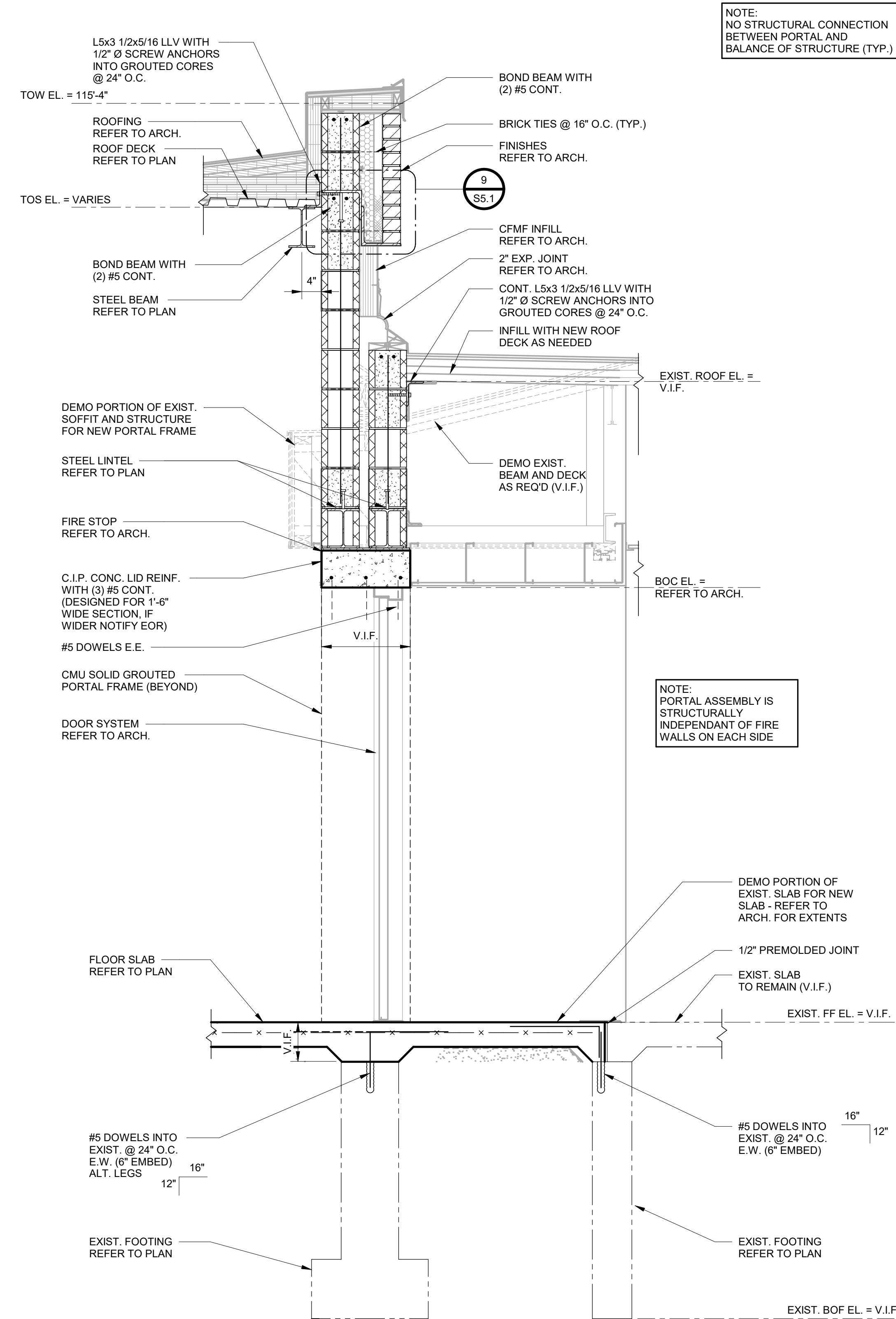
4 SECTION  
S7.2 3/4" = 1'-0"



3 SECTION  
S7.2 3/4" = 1'-0"



2 SECTION  
S7.2 3/4" = 1'-0"



1 SECTION  
S7.2 3/4" = 1'-0"



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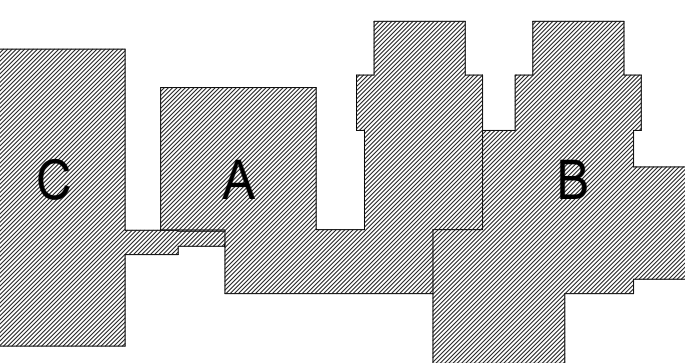


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Plymouth Canton Community Schools

DRAWING TITLE  
**SECTIONS AND DETAILS**



Key Plan: NTS

ISSUE DATES

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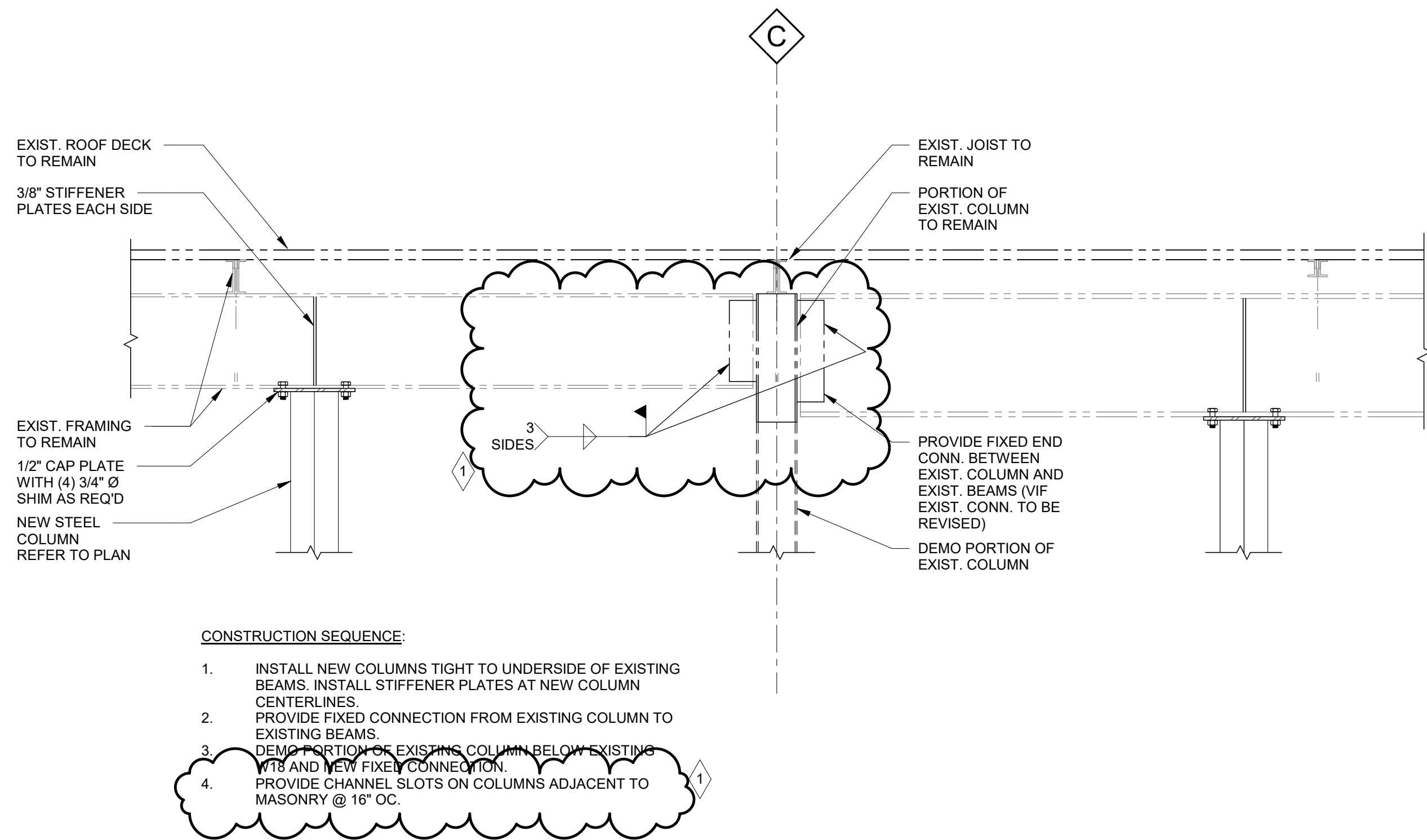
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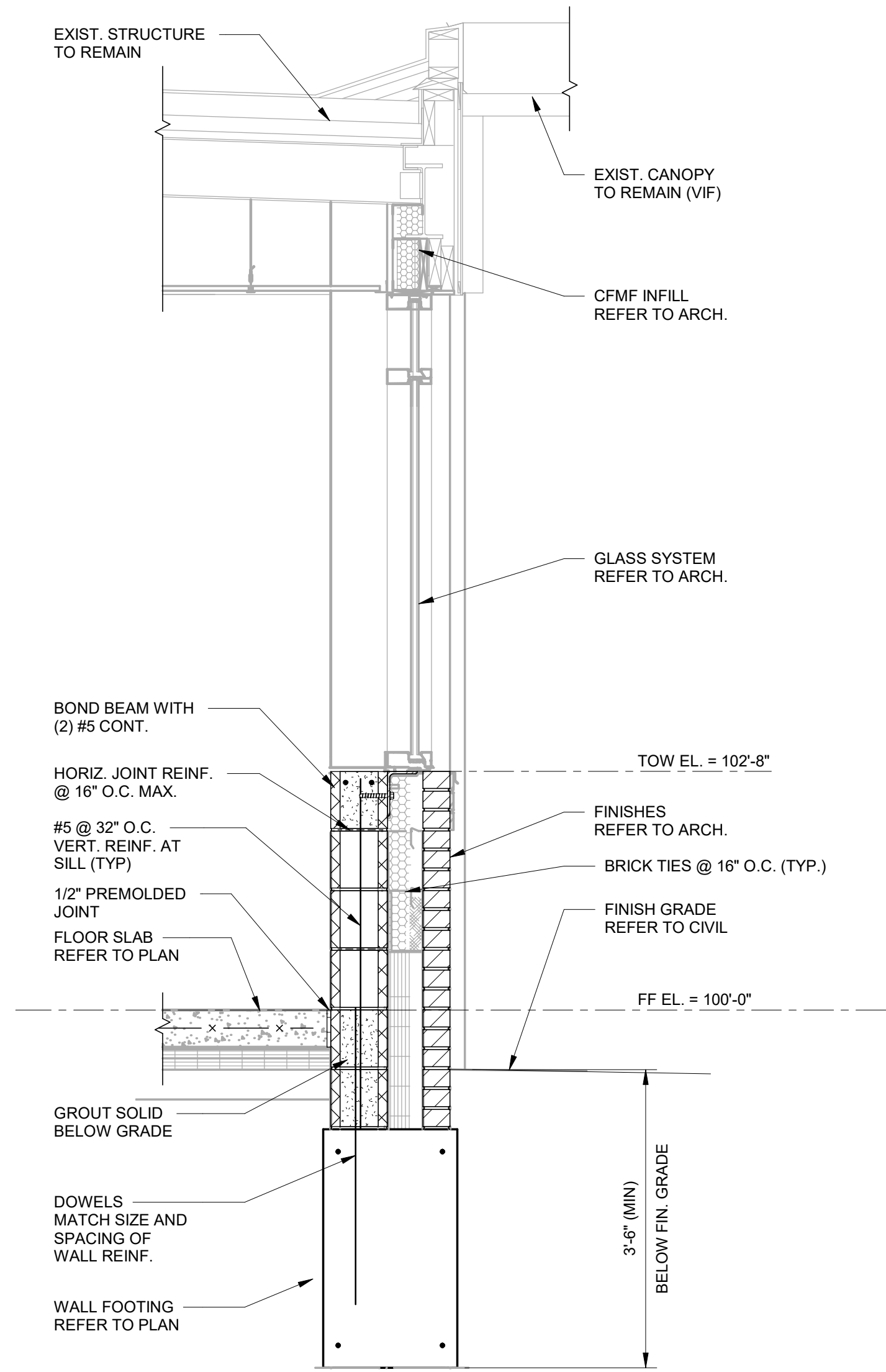
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**S7.2**





**2 SECTION**  
S7.3 3/4" = 1'-0"



**1 SECTION**  
S7.3 3/4" = 1'-0"



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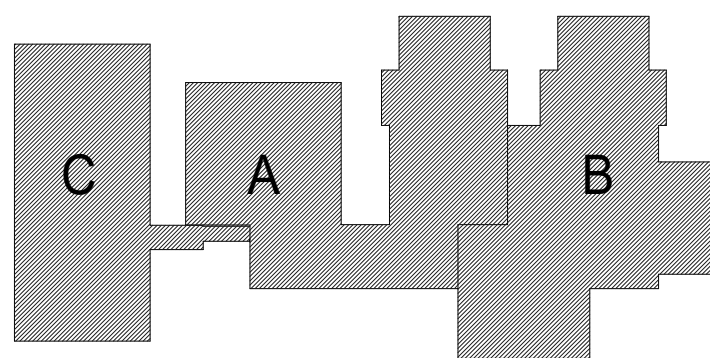
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Key Plan: NTS

ISSUE DATES

02-20-2023 ADDENDUM No 1  
01-30-2023 CONSTRUCTION DOCUMENTS

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DRAWING NO.

**S7.3**